

THEME ONE

UNIT

1

Number Sense and Operations

Decimal Place Value and Computation

► Concept 1 :

Decimals to the Thousandths Place

► Concept 2 :

Adding and Subtracting Decimals



CONCEPT 1

Decimals to the Thousandths Place

► Lesson 1

Decimals to the Thousandths Place

Learning Objectives:

- Students will read decimal numbers to the Thousandths place.
- Students will write decimal numbers to the Thousandths place.

► Lessons 2&3

- Place Value Shuffle
- Composing and Decomposing Decimals

Learning Objectives:

- Students will explain how a digit changes value as it moves to the left or right in a decimal or whole number.
- Students will compose and decompose decimals in multiple ways.

► Lesson 4

Comparing Decimals

Learning Objectives:

- Students will compare decimals to the Thousandths place.

► Lesson 5

Rounding Decimals

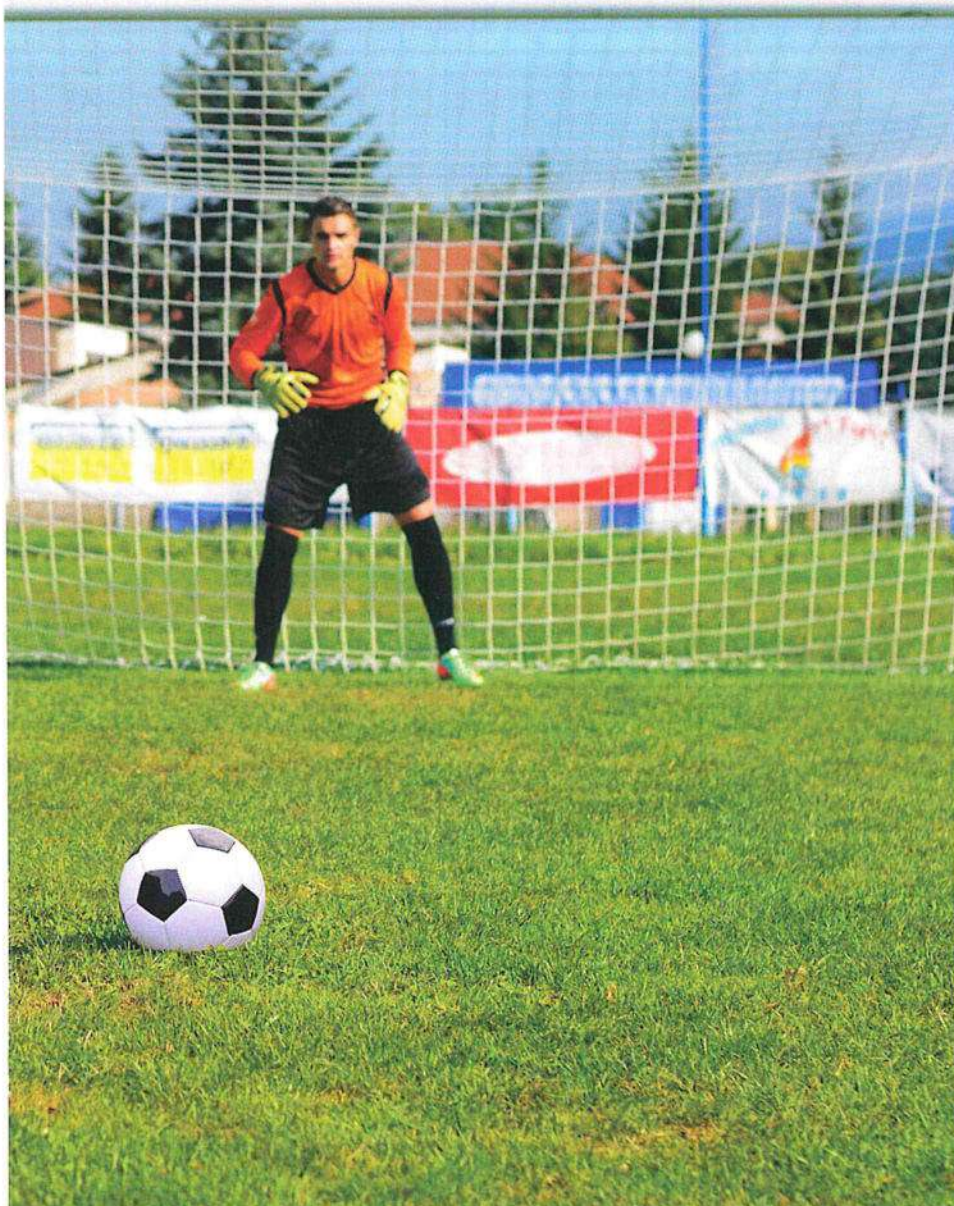
Learning Objectives:

- Students will round decimals to the nearest Tenth, Hundredth, or Thousandth.

Fast Fact

Each goal in a football game consists of two upright posts and joined at the top by a horizontal crossbar.

The distance between the posts is **7.32** m and the distance from the lower edge of the crossbar to the ground is **2.44** m.



Lesson

1

► Decimals to the Thousandths Place

Learn

Decimals to the Thousandths place

- A **decimal** is a number that uses a decimal point as **9.58**
- A decimal has one or more digits to the right of a decimal point.



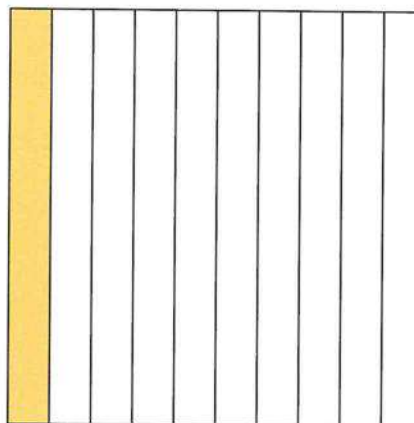
Did You Know?!

In 2009, Usain Bolt set the world record in the 100-meter sprint at **9.58** seconds. He still known as the fastest man in the world.

- You can use the following grid to illustrate the meaning of thousandth.

One whole is divided into

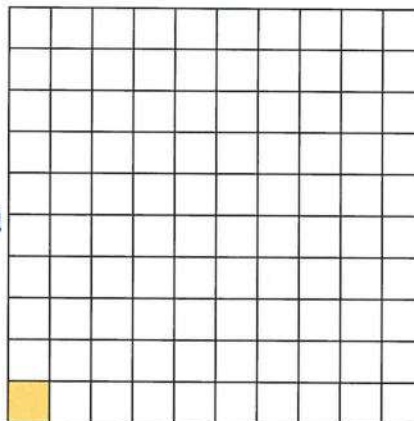
10 equal parts



The colored part
= 1 tenth
= $\frac{1}{10}$ (0.1)

or

100 equal parts



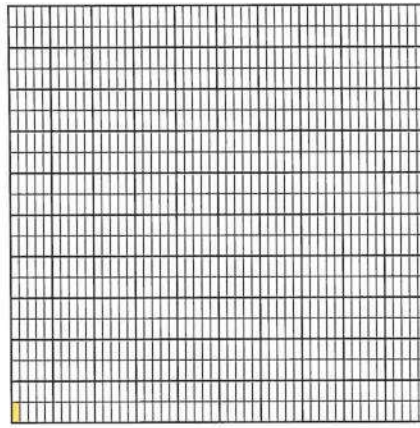
The colored part
= 1 hundredth
= $\frac{1}{100}$ (0.01)

Notes for parents :

- Let your child review place value from the Millions place to the Hundredth place.

or

1,000 equal parts



The colored part = 1 thousandth

$$= \frac{1}{1,000} (0.001)$$

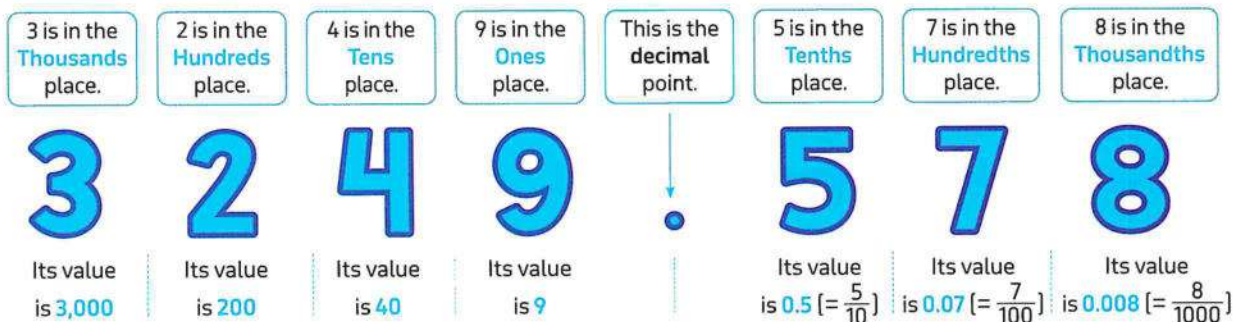
Note that

- Each tenth could be divided into 100 equal parts, each part represents one thousandth.
- Each hundredth could be divided into 10 equal parts, each part represents one thousandth.

• The value of each digit in any number depends on its place in this number

For Example :

Notice the value of each digit in the number 3,249.578



You can use the large place-value chart to help you read and write decimals as follows :

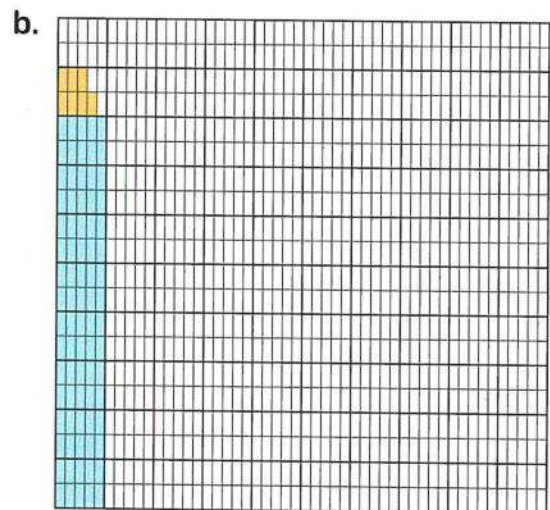
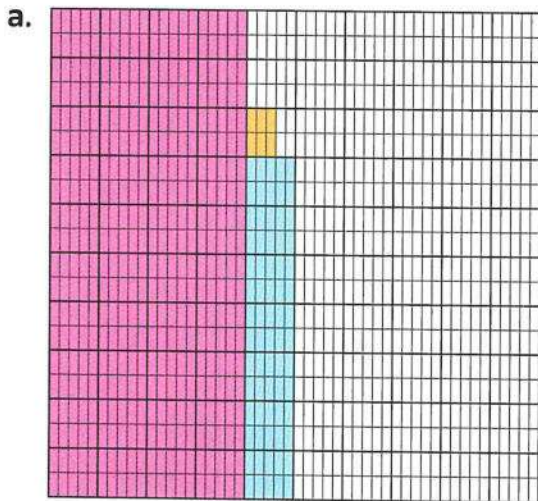
Milliards			Millions			Thousands			Ones			.	Decimals		
O	H	T	O	H	T	O	H	T	O	.	Tenths	Hundredths	Thousandths		
						3	2	4	9	.	5	7	8		

Standard Form : 3,249.578**Word Form :** Three thousand ,two hundred forty-nine and five hundred seventy – eight thousandths.**Unit Form :** 3 Thousands ,2 Hundreds ,4 Tens ,9 Ones ,5 Tenths ,7 Hundredths , 8 Thousandths.

- Help your child read numbers from the Milliards place to the Thousandths place.

Example 1

Record what decimal is shown :

**Solution**

a. 4 Tenths , 7 Hundredths , 6 Thousandths
 = 476 thousandths
 = $0.476 \left(\frac{476}{1,000} \right)$
 "Four hundred seventy-six thousandths"

b. 8 Hundredths, 7 Thousandths
 = 87 thousandths
 = $0.087 \left(\frac{87}{1,000} \right)$
 "Eighty - Seven thousandths"

Example 2

Write each of the following in word form.

a. 305.183

b. 84.005

c. 3,024.8

Solution

- a. Three hundred five and one hundred eighty-three thousandths.
 b. Eighty-four and five thousandths.
 c. Three thousand, twenty-four and eight tenths.

Example 3

In the number 6,354.792

- a. What is the value of 6 ?
 b. What is the value of 2 ?
 c. What does the digit 4 represent ?
 d. What is the value of the digit in the Hundredth place ?

Solution

a. 6,000

b. 0.002

c. 4 Ones

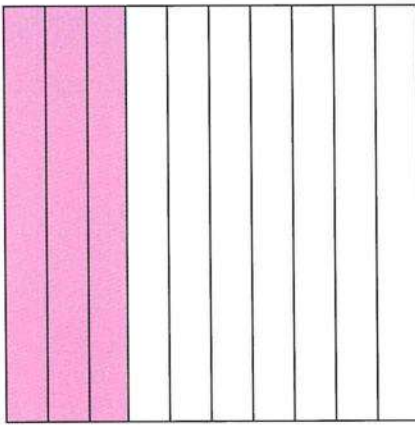
d. 0.09

Notes for parents :

- Help your child read and write decimal numbers to the Thousandths place.

Remark

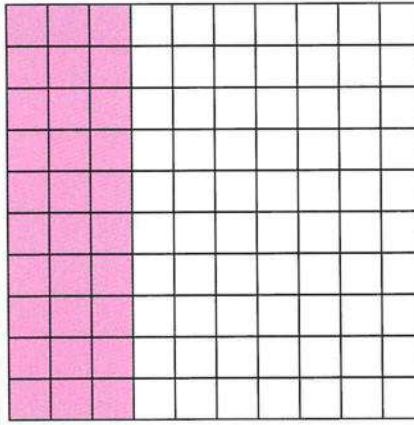
You can name the same amount in different ways as follows :



$$\frac{3}{10}$$

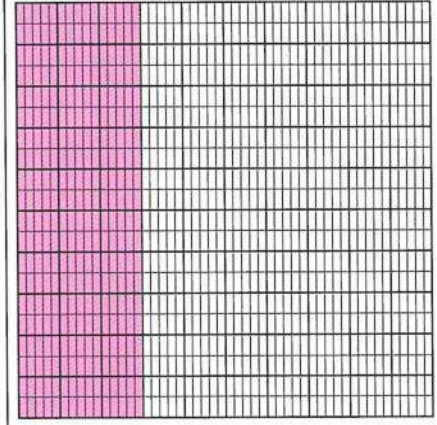
$$0.3$$

=



$$\frac{30}{100}$$

$$0.30$$



$$\frac{300}{1,000}$$

$$0.300$$

We deduce that :

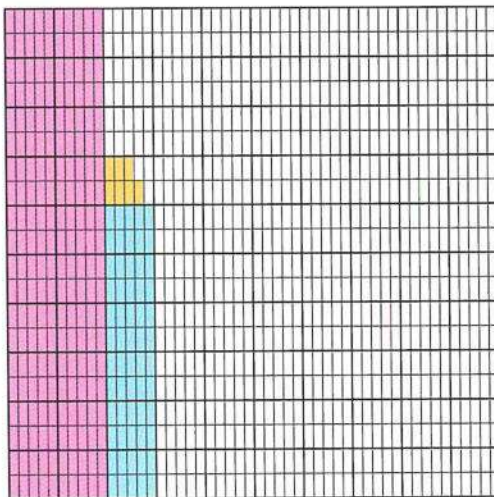
If you put zeroes after the last decimal digit in a number , then the value of this number doesn't change.



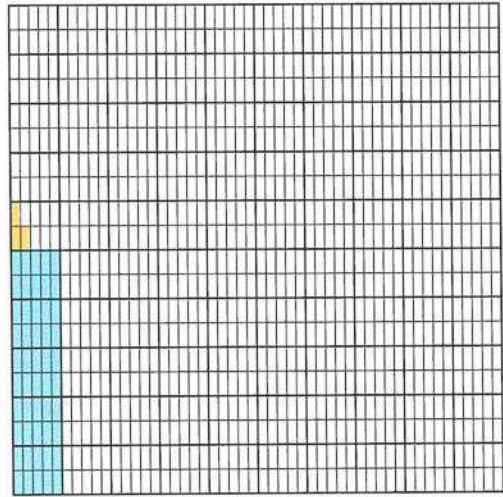
check your understanding

1. Record what decimal is shown :

a.



b.



2. Complete.

- In 942.358 , the digit 8 is in the _____ place. Its value is _____
- In 791.06 , the digit 0 is in the _____ place. Its value is _____
- In 302.91 , the digit 1 is in the _____ place. Its value is _____

Notes for parents :

- Give your child a decimal like 0.8 and ask him/her to name this decimal in different ways.

Exercise

1

on lesson 1

► Decimals to the Thousandths Place

REMEMBER

UNDERSTAND

APPLY

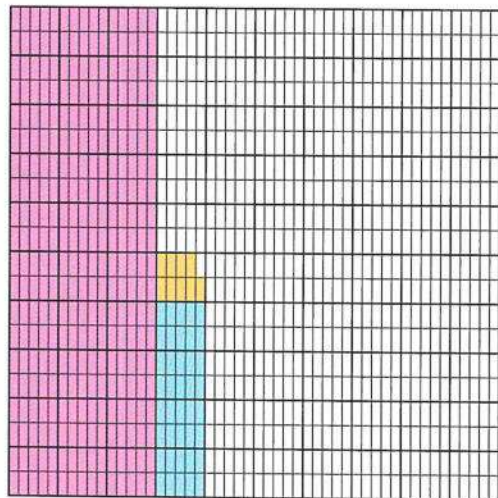
PROBLEM SOLVING

From the school book

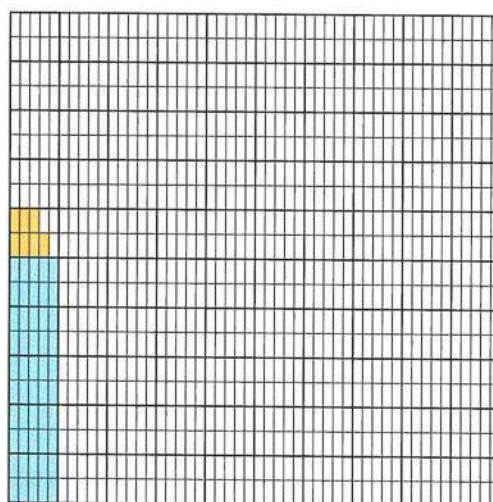
1. Record what decimal is shown.

●

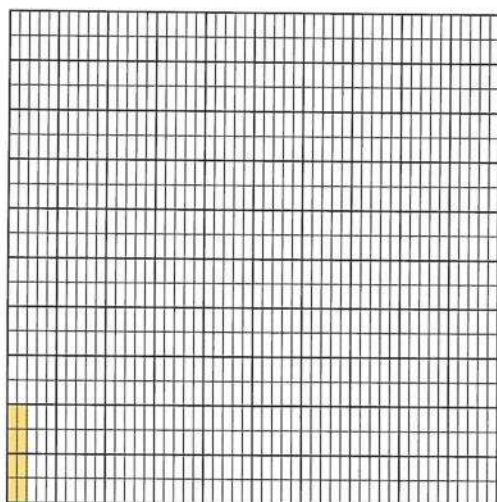
a.



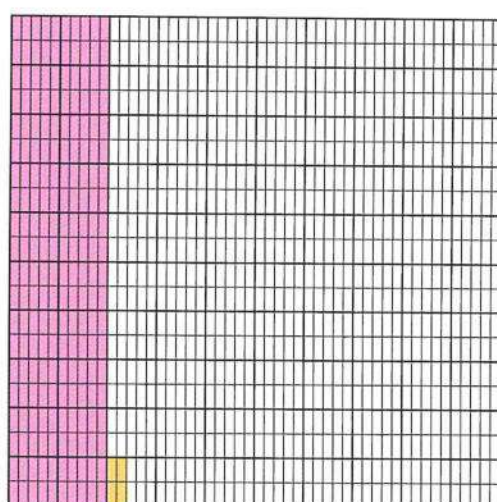
b.



c.



d.



2. Write each of the following in word form.

●

a. 504.21 _____

b. 4.231 _____

c. 49.08 _____

d. 0.534 _____

- e. 4,030.7 _____
- f. 4.029 _____
- g. 17.107 _____
- h. 1.802 _____
- i. 0.608 _____
- j. 8.002 _____

[Cairo - El Basateen & El Salam 24]

3. Write each of the following in standard form.

- | | |
|---|---|
| a. 24 hundredths _____ | b. 35 thousandths _____ |
| c. 8 Thousandths _____ | d. 7 and 14 thousandths _____ |
| e. 4 and 4 thousandths _____ | f. 1 and 5 tenths _____ |
| g. 9 and 700 thousandths _____ | h. 20 and 40 thousandths _____ |
| i. 7 thousand and 48 hundredths _____ | j. 3 million and 142 thousandths _____ |
| k. 2 milliard and 3 thousandths _____ | l. 4 Tenths, 8 Thousandths _____ |
| m. 5 Ones, 2 Thousandths _____ | |
| n. Two hundred thirty - five thousandths _____ | |
| | [Ismailia - El Kasaseen 24, Kafr El Sheikh - Bayala 24] |
| o. Three and twenty-five thousandths _____ | [El Beheira 23] |
| p. Two and one hundred nine thousandths _____ | |
| q. Fifteen and fifteen thousandths _____ | [Alexandria - El Montaza 24] |
| r. Thirty - six and twenty - five thousandths _____ | [Souhag - Akhmem 24] |
| s. Four and three hundredths _____ | [Cairo - El Maadi 24] |

4. In the number 729.458

- | | |
|--|---|
| a. What is the value of 4 ?
_____ | b. What is the value of 8 ?
_____ |
| c. What does the digit 2 represent ?
_____ | d. What does the digit 9 represent ?
_____ |
| e. What is the value of the digit in the Hundredths place ?
_____ | |

5. Complete.

- a. 5 Tenths = _____ hundredths. [Cairo - El Mokattam 24, Kafr El Sheikh - Bayala 24]
- b. The value of the digit 6 in the number 2.612 is _____. [Giza - Abo El Nomrous 23]
- c. The place value of the digit 5 in the number 3.514 is _____. [Cairo - West 24, Souhag - Tima 24]
- d. In 35.627, the digit 7 represents _____. [Giza - Awseem 24]
- e. The place value of the digit 4 in 61.24 is _____. [Souhag - Girga 24]
- f. The value of the digit 5 in 3.215 is _____. [El Menia - Bani Mazar 24]
- g. The value of the digit "0" in the number 16.205 is _____. [Giza - El Omraniya 24]
- h. The value of the digit 2 in the number 5.264 is _____. [El Beheira - Rasheed 24]
- i. The value of the digit 9 in the decimal number 91.85 is equal to _____. [Cairo - Nasr City 24]

6. How many whole numbers, tenths, hundredths and thousandths does the number 0.007 have?

7.  The Purple Heron is tall at 70 to 90 centimeters, but it weighs only 0.50 to 1.35 kilograms. Below are the weights of three Purple Herons.

For each number, record the following :

- a. The digit that is in the Tenths place.
- b. The digit that is in the Ones place.
- c. The digit that is in the Hundredths place.

Bird One	0.65 kilogram
Bird Two	1.27 kilograms
Bird Three	0.875 kilogram

8. Math around Egypt : Gas Price Decimals

Look at the list of different petrol prices in Egypt.

- a. Which type of petrol is the least expensive ?
- b. Which type of petrol is the most expensive ?

Gas Prices per Liter, April 2021

80 Octane petrol : 6.75 L.E.

92 Octane petrol : 8.00 L.E.

95 Octane petrol : 9.00 L.E.

Multiple Choice Questions

Choose the correct answer.

1. $\frac{555}{1,000} = \underline{\hspace{2cm}}$

[El Menia - Deir Mawas 24]

- A. 555 B. 5.55
C. 55.5 D. 0.555

2. $\frac{75}{100} = \underline{\hspace{2cm}}$

[Port Said, Port Fuad 24]

- A. 7.5 B. 75.100
C. 0.75 D. 7.05

3. Three and seventy-five hundredths = $\underline{\hspace{2cm}}$

[El Beheira - Housh Essa 23, El Menia - Matai 24]

- A. 3.57 B. 3.75
C. 375 D. 35.7

4. "Twenty - seven and sixty - six thousandths" in standard form is $\underline{\hspace{2cm}}$

[El Menia - Samalut 24]

- A. 27.66 B. 66.27
C. 27.066 D. 270.66

5. 2 Tenths , 5 Hundredths = $\underline{\hspace{2cm}}$

[Cairo - Al Sayeda Zeinab 24]

- A. 0.205 B. 0.25
C. 0.025 D. 0.52

6. 71 tenths = $\underline{\hspace{2cm}}$

[Kafr El Sheikh - Bayala 24, El Monofia, Sers El Laian 24]

- A. 0.71 B. 71
C. 71 D. 710

7. 1,234 hundredths = $\underline{\hspace{2cm}}$

[Alexandria - Montaza 24]

- A. 1.234 B. 12.34
C. 123.4 D. 0.1234

8. The value of 4 in the number 3.124 is $\underline{\hspace{2cm}}$

[El Menofia - Ashmoon 24]

- A. 4 B. 0.4
C. 0.04 D. 0.004

9. The value of the digit 5 in the number 3.514 is $\underline{\hspace{2cm}}$

[Luxor 24, Port Said 24]

- A. 5 B. 0.5
C. 0.05 D. 0.005

10. The value of the digit 3 in the number 14.239 is $\underline{\hspace{2cm}}$

[El Fayoum 24]

- A. 30 B. 0.3
C. 0.03 D. 0.003

11. The place value of the digit 3 in the number 15.32 is $\underline{\hspace{2cm}}$

[Aswan 23]

- A. Ones. B. Hundreds.
C. Tenths. D. Thousandths.

12. Which of the following doesn't equal four hundred thousandths ?

- A. 0.004 B. 0.40
C. 0.4 D. 0.400

13. The decimal fraction 0.053 reads $\underline{\hspace{2cm}}$

[Cairo - Al Khalifa and Al Mokattam 23]

- A. fifty-three hundredths.
B. fifty-three hundreds.
C. thirty-five hundredths.
D. fifty-three thousandths.

14. $0.300 = \underline{\hspace{2cm}}$

[Alexandria - Agmi 24]

- A. 3 Tenths B. $\frac{300}{100}$
C. $\frac{30}{10}$ D. $\frac{3}{100}$

Lessons 2 & 3

- Place Value Shuffle
- Composing and Decomposing Decimals

Learn 1 Place value shuffle

- 1** If a whole number or a decimal is multiplied by [10], then each digit from this number shifts to the left one spot on the place-value chart and the value of each digit increases ten times.

For Example: 714×10

Millions			Thousands			Ones			.	Decimals	
H	T	O	H	T	O	H	T	O	.	Tenths	Hundredths
						7	1	4	.	0	0
						7	1	4	.	0	0

- Record 714 on the place-value chart.
- Shift each digit to the left one spot to get the number "7,140"
- Then $714 \times 10 = 7,140$
- The value of the whole number "714" increased when multiplying by 10
- The value of 7 increased when multiplying by 10 from 700 to 7,000
- The value of 1 increased when multiplying by 10 from 10 to 100
- The value of 4 increased when multiplying by 10 from 4 to 40

Another Example: 7.14×100

Millions			Thousands			Ones			.	Decimals	
H	T	O	H	T	O	H	T	O	.	Tenths	Hundredths
								7	.	1	4
						7	1	4	.	4	0
						7	1	4	.	0	0

Note that

When multiplying by [100] each digit shifts to the left two spots, then the value of each digit increases 100 times.

- Then $7.14 \times 100 = 714$

Notes for parents :

- Let your child explain how a digit changes value as it moves to left in a decimal or a whole number.

- 2** If a whole number or a decimal is divided by [10], then each digit from this number shifts to the right one spot on the place-value chart and the value of each digit decreases ten times.

For Example: $615 \div 10$

Millions			Thousands			Ones			.	Decimals	
H	T	O	H	T	O	H	T	O	.	Tenths	Hundredths
						6	1	5	.		
							6	1	.	5	

- Record 615 on the place-value chart.
- Shift each digit to the right one spot to get the number 61.5
- Then $615 \div 10 = 61.5$
- The value of the whole number "615" decreased when dividing by 10
- The value of 6 decreased when dividing by 10 from 600 to 60
- The value of 1 decreased when dividing by 10 from 10 to 1
- The value of 5 decreased when dividing by 10 from 5 to 0.5

Another Example: $61.5 \div 100$

Ones			.	Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths
	6	1	.	5		
		6	.	1	5	
		0	.	6	1	5

Note that

When dividing by [100] each digit shifts to the right two spots, then the value of each digit decreases 100 times.

- Then $61.5 \div 100 = 0.615$

Remark

Dividing any number by 10 is the same as multiplying this number by $\frac{1}{10}$
 So, $362 \div 10 = 362 \times \frac{1}{10}$

- Let your child explain how a digit changes value as it moves to the right in a decimal or a whole number.

Example 1

Use the place-value charts to solve each problem.

a. 8.7×10

b. $1.35 \times 1,000$

c. $2.5 \div 10$

d. $6.2 \div 100$

Solution 

a.

Thousands			Ones			.	Decimals		
H	T	O	H	T	O	.	Tenths	Hundredths	Thousandths
					8	.	7		
			8	7		.	0		

$$8.7 \times 10 = 87$$

b.

Thousands			Ones			.	Decimals		
H	T	O	H	T	O	.	Tenths	Hundredths	Thousandths
					1	.	3	5	
				1	3	.	5	0	
			1	3	5	.	0	0	
		1	3	5	0	.	0	0	

$$1.35 \times 1,000 = 1,350$$

c.

Thousands			Ones			.	Decimals		
H	T	O	H	T	O	.	Tenths	Hundredths	Thousandths
					2	.	5		
					0	.	2	5	

$$2.5 \div 10 = 0.25$$

d.

Thousands			Ones			.	Decimals		
H	T	O	H	T	O	.	Tenths	Hundredths	Thousandths
					6	.	2		
						.	6	2	
					0	.	0	6	2

$$6.2 \div 100 = 0.062$$

Notes for parents :

- Help your child solve more problems on multiplying and dividing by 10 or 100.

 check your understanding

Use the place-value charts to solve each problem. Fill in the blanks to show how the value of each digit also changed.

a. 85×10

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
		—	—	.	—	—
		—	—	.	—	—

- The value of the whole number _____ [increased / decreased]
- The value of 5 [increased / decreased] when multiplying by 10 from _____ to _____
- The value of 8 [increased / decreased] when multiplying by 10 from _____ to _____

b. $942 \div 100$

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
	—	—	—	.	—	—
	—	—	—	.	—	—

- The value of the whole number _____ [increased / decreased]
- The value of 9 [increased / decreased] when dividing by 100 from _____ to _____
- The value of 2 [increased / decreased] when dividing by 100 from _____ to _____

c. 6.31×100

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
		—	—	.	—	—
		—	—	.	—	—

- The value of the whole number _____ [increased / decreased]
- The value of 3 [increased / decreased] when multiplying by 100 from _____ to _____
- The value of 1 [increased / decreased] when multiplying by 100 from _____ to _____

d. $74.8 \div 10$

Thousands		Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths	
	—	—	—	.	—	—	
	—	—	—	.	—	—	

- The value of the whole number _____ [increased / decreased]
- The value of 7 [increased / decreased] when dividing by 10 from _____ to _____
- The value of 8 [increased / decreased] when dividing by 10 from _____ to _____

• Let your child discover how the decimal point moves when multiplying and dividing by 10 or 100

Learn 2 Composing and decomposing decimals

- **Composing** decimals means [put together]
- **Decomposing** decimals means [broken apart]
- You can decompose 843.572 in different ways using place-value chart :

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths
	8	4	3	.	5	7	2

► 1st way [expanded form] :

$$843.572 = 800 + 40 + 3 + 0.5 + 0.07 + 0.002$$

► 2nd way :

$$843.572 = 843 + 0.572$$

► 3rd way :

$$843.572 = 843 + 0.5 + 0.07 + 0.002$$

There are many answers that equal 843.572 when composed.



Example 2

Record the number 504.82 in the place-value chart and decompose this number in expanded form then decompose it in two other ways.

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths
	—	—	—	.	—	—	—

- 1st way [expanded form] : _____
- 2nd way : _____
- 3rd way : _____

Solution

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths
	5	0	4	.	8	2	

- 1st way [expanded form]: $504.82 = 500 + 4 + 0.8 + 0.02$
- 2nd way : $504.82 = 500 + 4 + 0.82$
- 3rd way : $504.82 = 504 + 0.8 + 0.02$

You can choose any other answers.

Notes for parents :

- Let your child begin by reviewing how to write number in expanded form and learn that number can be decomposed in many different ways.

Example 3

Decompose the following numerals using expanded form.

- a. 640.078
b. Twenty-three and forty-two thousandths.

Solution 

- a. $640.078 = 600 + 40 + 0.07 + 0.008$
b. $23.042 = 20 + 3 + 0.04 + 0.002$

**Example 4**

Compose each of the following.

- a. $4,000 + 80 + 7 + \frac{1}{10} + 0.002$
b. $420 + 0.2 + 0.07 + 0.009$
c. $5,900 + 0.3 + \frac{8}{1,000} + 70 + 2$

Solution 

- a. 4,087.102 b. 420.279 c. 5,972.308



check your understanding

Complete the following.

1. Compose : 452.087

Decompose : _____

2. Compose : 204.005

Decompose : _____

3. Compose : _____

Decompose : $540 + 0.2 + 3 + 0.007 + \frac{9}{100}$

• Help your child compose and decompose decimals in multiple ways.

Exercise

2

on lessons 2&3

► Place Value Shuffle

► Composing and Decomposing Decimals

● REMEMBER

● UNDERSTAND

● APPLY

● PROBLEM SOLVING

From the school book

Place value shuffle

1. Use the place-value charts to solve each problem. Fill in the blanks to show how the value of each digit also changed.

a. $85 \times 10 =$ _____

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
				.		
				.		

- The value of the whole number _____ [increased/decreased] when multiplying by 10
- The value of the _____ [first digit] _____ [increased/decreased] when multiplying by 10 from _____ to _____
- The value of the _____ [second digit] _____ [increased/decreased] when multiplying by 10 from _____ to _____

b. $57 \div 10 =$ _____

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
				.		
				.		

- The value of the whole number _____ [increased/decreased] when dividing by 10
- The value of the _____ [first digit] _____ [increased/decreased] when dividing by 10 from _____ to _____
- The value of the _____ [second digit] _____ [increased/decreased] when dividing by 10 from _____ to _____

c. $6.5 \times 10 =$ _____

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
				.		
				.		

- The value of the whole number _____ [increased/decreased] when multiplying by 10
- The value of the _____ [first digit] _____ [increased/decreased] when multiplying by 10 from _____ to _____
- The value of the _____ [second digit] _____ [increased/decreased] when multiplying by 10 from _____ to _____

d. $7.3 \times 100 =$ _____

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
				.		
				.		

- The value of the whole number _____ [increased/decreased] when multiplying by 100
- The value of the _____ [first digit] _____ [increased/decreased] when multiplying by 100 from _____ to _____
- The value of the _____ [second digit] _____ [increased/decreased] when multiplying by 100 from _____ to _____

e. $345 \div 10 =$ _____

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths
				.		
				.		

- The value of the whole number _____ [increased/decreased] when dividing by 10
- The value of the _____ [first digit] _____ [increased/decreased] when dividing by 10 from _____ to _____
- The value of the _____ [second digit] _____ [increased/decreased] when dividing by 10 from _____ to _____
- The value of the _____ [third digit] _____ [increased/decreased] when dividing by 10 from _____ to _____

2. Form the place-value chart to solve each problem.

a. $75 \times 10 = \underline{\hspace{2cm}}$

b. $43 \times 100 = \underline{\hspace{2cm}}$

c. $2.5 \times 10 = \underline{\hspace{2cm}}$

d. $14.52 \times 10 = \underline{\hspace{2cm}}$

e. $218 \div 10 = \underline{\hspace{2cm}}$

f. $4.9 \div 10 = \underline{\hspace{2cm}}$

g. In the problem $74.8 \div 10$. The value of the digit 4 decreased from 4 to $\underline{\hspace{2cm}}$

[El Monofia - Sers El Laian 24]

Composing and Decomposing Decimals

3. In the following problem, record the number in the place-value chart and decompose this number in expanded form and then in two other ways.

a.  34.527

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths

• 1st way [expanded form] : $\underline{\hspace{4cm}}$

• 2nd way : $\underline{\hspace{4cm}}$

• 3rd way : $\underline{\hspace{4cm}}$

b.  21.045

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths

• 1st way [expanded form] : $\underline{\hspace{4cm}}$

• 2nd way : $\underline{\hspace{4cm}}$

• 3rd way : $\underline{\hspace{4cm}}$

c. 302.504

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths

• 1st way [expanded form] : $\underline{\hspace{4cm}}$

• 2nd way : $\underline{\hspace{4cm}}$

• 3rd way : $\underline{\hspace{4cm}}$

d. 231.128

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths

- 1st way [expanded form] : _____
- 2nd way : _____
- 3rd way : _____

e. 508.17

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths

- 1st way [expanded form] : _____
- 2nd way : _____
- 3rd way : _____

4. Write each of the following in standard form.

- a. $5 + 0.3 + 0.01 + 0.007 =$ _____ [Giza - Awseem 24]
- b. $10 + 3 + 0.2 + 0.06 =$ _____ [Alexandria - First Montaza 23]
- c. $10 + 8 + 0.3 + 0.009 =$ _____ [El Monofia - Shibben El Kom 23]
- d. $8 + 0.2 + \frac{6}{100} + 0.009 =$ _____ [Cairo 23]
- e. $2 + 0.9 + \frac{8}{100} + \frac{2}{1,000} =$ _____ [El Menia - Deir Mowas 23]
- f. $5,000 + 40 + 9 + 0.2 + 0.007 =$ _____
- g. $700 + 0.4 + 0.009 =$ _____
- h. $9 + 0.003 + 0.5 + 10 =$ _____ [Cairo - El Maadi 24]
- i. $40 + 0.8 + 0.009 + 500 =$ _____
- j. $0.003 + 0.2 + 0.01 + 91,000 =$ _____ [Cairo - El Sherouk 23]
- k. $6,000 + 70,000 + 0.2 + 4 + 0.09 + 0.005 =$ _____
- l. $70 + 8,000 + 0.009 + 0.1 + 3 =$ _____

5. Decompose each of the following in expanded form.

- a. $59.784 = \underline{\hspace{2cm}}$ [Cairo - El Sahel 24]
 b. $800.57 = \underline{\hspace{2cm}}$ [Luxor 24]
 c. $156,327.194 = \underline{\hspace{2cm}}$ [Aswan 23]
 d. Two and forty-one thousandths = $\underline{\hspace{2cm}}$
 e. Seventy-nine thousandths = $\underline{\hspace{2cm}}$
 f. 8 tens, 4 ones, 3 tenths, 6 hundredths, 9 thousandths = $\underline{\hspace{2cm}}$
 g. 4 hundreds, 7 hundredths, 8 thousandths = $\underline{\hspace{2cm}}$



6. Complete each of the following.

- a. $4.208 = \underline{\hspace{2cm}} + 0.2 + 0.008$
 b. $70.106 = 70 + 0.1 + \underline{\hspace{2cm}}$
 c. $32.308 = \underline{\hspace{2cm}} + 0.3 + 2 + 30$ [Ismailia - Fayed 24]
 d. $\underline{\hspace{2cm}} = 120 + 0.204$
 e. $\underline{\hspace{2cm}} = 4 + 0.005 + 0.3$
 f. $34.012 = 34 + \underline{\hspace{2cm}}$
 g. $\underline{\hspace{2cm}} = 4 + 30 + 400 + 0.008 + 0.02$
 h. Seventy and eight thousandths = $\underline{\hspace{2cm}} + \underline{\hspace{2cm}}$
 i. Sixteen and seven tenths = $\underline{\hspace{2cm}} + \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$
 j. 283 thousandths = $\underline{\hspace{2cm}} + 0.2 + 0.08$
 k. 57 thousandths = $0.007 + \underline{\hspace{2cm}}$
 l. 15.7 tenths = $1 + \underline{\hspace{2cm}} + 0.07$
 m. 1,482 hundredths = $14 + \underline{\hspace{2cm}}$





Multiple Choice Questions

Choose the correct answer.

1. $4.7 \times 100 =$ _____

[El Monofia - Menof 24, Sers El Laian 24]

- A. 47 B. 470
C. 0.047 D. 0.47

2. $5.26 \times 100 =$ _____

[Ismailia 23]

- A. 5,260 B. 0.526
C. 526 D. 52.6

3. $75 \bigcirc 7.5 \times 10$

[Alexandria - El Gamarek 24]

- A. < B. >
C. =

4. When dividing 316 by 10, then the value of 6 becomes _____

[El Monofia - Ashmoon 24]

- A. 0.6 B. 60
C. 0.06 D. 600

5. 5,000 not equals _____

[El Menia - Mallawi 24, Giza - Awseem 23]

- A. $5 \times 1,000$ B. 50×100
C. 500×10 D. 500×100

6. $28.4 \div$ _____ $= 2.84$

[Aswan - Kom Ombo 23]

- A. 10 B. 100
C. 1,000 D. 10,000

7. $0.008 + 0.07 + 20 =$ _____

[Cairo - EL Sharouk 24, Rod El Farg 24]

- A. 20.807 B. 20.078
C. 20.78 D. 87.02

8. 6 Ones + 5 Tenths + 7 Thousandths = _____

[Aswan 23]

- A. 0.756 B. 6.507
C. 657 D. 6,507

9. The standard form of $1 + 0.7 + 0.07$ is _____

[El Fayoum 24, Cairo - West 24, Alex. - El Montaza 24]

- A. 1.71 B. 1.77
C. 77.1 D. 17.7

10. $35.605 = 35 +$ _____

[Giza - Omrania 24]

- A. 605 B. 0.65
C. 0.605 D. 0.05

11. $0.2 +$ _____ $= 7.2$

[El Kalyoubia 23]

- A. 7 B. 0.7
C. 70 D. 0.07

12. The number fifteen and fifteen hundredths in expanded form is _____

[Giza - El Haram 24]

- A. $10 + 5 + 0.1 + 0.005$
B. $10 + 5 + 0.05 + 0.001$
C. $10 + 5 + 0.1 + 0.05$
D. $10 + 5 + 0.01 + 0.005$

► Comparing Decimals

Learn How to compare two decimals

You can use place-value charts to compare decimals.

► **Examples :**

Compare 2.948 and 2.957

Ones				Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths
		2	.	9	4	8
		2	.	9	5	7

- Begin with the digit in the greatest place value.
- **Compare ones :** 2 ones = 2 ones
- **Compare tenths :** 9 tenths = 9 tenths
- **Compare hundredths :** 4 hundredths < 5 hundredths

So, $2.948 < 2.957$

Compare 0.26 and 0.206

Ones				Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths
		0	.	2	6	0
		0	.	2	0	6

- Begin with the digit in the greatest place value.
- **Compare ones :** 0 ones = 0 ones
- **Compare tenths :** 2 tenths = 2 tenths
- **Compare hundredths :** 6 hundredths > 0 hundredths

So, $0.26 > 0.206$

Example 1

Use place-value chart to compare the following decimals :

a. 52.008 and 52.8

b. 3.02 and 3.019

c. 67.5 and 67.500

Notes for parents :

- Remind your child to begin comparing with the greatest place value.



Solution

a.

Ones				Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths
	5	2	.	0	0	8
	5	2	.	8	0	0

$5 = 5$, $2 = 2$, $0 < 8$ Since, $0 < 8$

So, $52.008 < 52.8$

b.

Ones				Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths
		3	.	0	2	0
		3	.	0	1	9

$3 = 3$, $0 = 0$, $2 > 1$ Since, $2 > 1$

So, $3.02 > 3.019$

c.

Ones				Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths
	6	7	.	5	0	0
	6	7	.	5	0	0

$6 = 6$, $7 = 7$, $5 = 5$, $0 = 0$ $0 = 0$

So, $67.5 = 67.500$

Line up the decimal points.
Compare the digits,
beginning with the greatest
place value.

**Example 2**

Compare 2.135 and 2.137

Solution

To compare 2.135 and 2.137, follow the following steps :

Step 1	Step 2	Step 3	Step 4
Compare the ones. 2.135 ↓ 2.137 the same number of ones	Compare the tenths. 2.135 ↓ 2.137 the same number of tenths	Compare the hundredths. 2.135 ↓ 2.137 the same number of hundredths	Compare the thousandths. 2.135 ↓ 2.137 $5 < 7$

So, $2.135 < 2.137$

• Ask your child how is comparing decimals like comparing whole numbers.

Example 3

Compare using "<, > or =".

a. 0.395 $\frac{385}{1,000}$

b. 28 thousandths 0.28

c. 4 ones, 4 hundredths, 5 thousandths 4.054

Solution

a. Since, $\frac{385}{1,000} = 0.385$

So, $0.395 > 0.385$

b. Since, 28 thousandths = 0.028

So, $0.028 < 0.28$

c. Since, 4 ones, 4 hundredths, 5 thousandths = 4.045

So, $4.045 < 4.054$

 **Check** your understanding

Compare using ">, < or =".

a. 3.204 3.24

b. 19.2 19.200

c. 20.7 20.077

d. 1.01 1.099

e. 9.08 9.079

f. 14.010 $14\frac{9}{10}$

g. 4.12 $4 + 0.1 + 0.007$

h. 5 thousandths 0.500

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Notes for parents :

- Ask your child to explain the strategies he/she uses to compare decimals.

Exercise

3

on lesson 4

► Comparing Decimals

● REMEMBER ● UNDERSTAND ● APPLY ● PROBLEM SOLVING

From the school book

1. Rewrite the decimals in the chart. Use the symbols "> , < or =".

a. 4.08



4.8

Ones			.	Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths

b. 15.3



15.300

Ones			.	Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths

c. 230.03



230.009

Ones			.	Decimals		
H	T	O	.	Tenths	Hundredths	Thousandths

2. Compare the decimal numbers using the symbols "> , < or =".

Draw a place value chart to help you, if needed.

a. 0.2



0.193

b. 0.013



0.031

c. 0.007



0.07

d. 45.057



45.100

e. 0.10



0.100

f. 98.013



98.101

g. 50.009



50.100

h. 10.1



10.011

i. 34.56



3.456

j. 0.48



0.480

k. 2.01



2.099

l. 34.5



34.500

m. 87.3



87.03

n. 2.197



2.179

o. 2.19



2.190

p. 3.011



3.001

3.  At the Fayoum Basin, temperatures vary greatly. The numbers are the temperatures recorded on one day in May. All numbers are in degrees Celsius. Compare each set of numbers using the symbols "> , < or =".

a. 29.9° _____ 30.2°

c. 40.5° _____ 41.0°

e. 38.80° _____ 38.8°

b. 36.5° _____ 35.6°

d. 35.2° _____ 34.7°

4. Compare the numbers using "> , < or =".

a. 2.71 $2\frac{8}{100}$

c. 1.002 $\frac{1,002}{1,000}$

e. 4.000 $\frac{400}{1,000}$

g. 3 thousandths $\frac{30}{100}$

i. 8 tenths 0.799

k. 5.102 $5 + 0.1 + 0.02$

m. $8 + 0.009$ $8 + 0.1 + 0.001$

o. 7 ones , 5 thousandths 7.05

p. 2 ones , 3 tenths , 4 thousandths 2.34

q. 8.004 4 ones , 8 thousandths

r. $3\frac{4}{1,000}$ 3 ones , 4 hundredths

b. 2.007 $7\frac{2}{1,000}$

d. 16.24 $16\frac{224}{1,000}$

f. 99.257 1,234 tenths

h. Eighteen thousandths 0.02

j. 0.402 402 thousandths

l. 4.904 $4 + 0.9 + 0.004$

n. 407.05 $400 + 7 + 0.005$



5. Circle all the decimal numbers that are greater than 4.3

3.4 , 4.03 , 4.34 , 4.300 , 3.99 , 4.7 , 4.003

6. Circle all the decimal numbers that are smaller than 2.104

2.102 , 2.401 , 2.14 , 2.199 , 2.11 , 2.7 , 2.014

7.  Select the largest number :

1.401 , 1.341 , 1.440 , 1.055 , 1.3 , 1.30 , 1.28 , 1.49

8.  Select the smallest number :

20.09 , 20.1 , 20.001 , 20.011 , 20.10 , 20.010 , 20.9 , 20.21

9. Order from least to greatest.

a. 1.401 , 1.055 , 1.3 , 1.28

[Cairo - Heliopolis 23]

b. 1.662 , 1.616 , 1.661 , 1.166

c. 0.096 , 2.56 , 1.26 , 0.27

[El Menia 23]

d. 2.547 , 9.258 , 1.253 , 4.325

[Cairo - Helwan 24]

e. 0.004 , 0.071 , 0.7 , 0.11 , 0.05

[El Monofia - Shebin El Koum 24]

f. 80.21 , 80.012 , 8.102 , 8.012 , 80.09

10. Youssef ran 2.2 kilometers during track practice and

Nader ran 2.099 kilometers.

Who ran the greater distance ?



11.  Give an example of two decimal numbers where the number with more decimal digits is smaller than the other number.

12.  Give an example of two decimal numbers where the number with more decimal digits is equal to the other number.

Multiple Choice Questions

Choose the correct answer.

1. 1.27 ☐ 1.72 [Giza - Abo El Nomrous 24]

- A. $<$ B. $>$
C. $=$ D. $\geq$

2. 25.12 ☐ 25.056 [El Beheira 23, Cairo 24]

- A. $>$ B. $<$
C. $=$ D. $\leq$

3. 36.5 ☐ 35.6 [Port Said 23]

- A. $<$ B. $>$
C. $=$ D. Others

4. 456.25 ☐ 45.625 [Kafir El Sheikh 24]

- A. $<$ B. $=$
C. $>$

5. $5.36 > \text{_____}$
[Cairo - Al Khalifa and Al Mokattam 23]

- A. 5.37 B. 5.362
C. 5.366 D. 3.561

6. $7.54 < \text{_____}$ [Alexandria - West 24]

- A. 7.145 B. 7.216
C. 7.6 D. 7.399

7. 5 Tenths ☐ 0.099 [Giza 24]

- A. $>$ B. $<$
C. $=$ D. $\leq$

8. 19 hundredths ☐ 19 thousandths

- A. $>$ B. $<$
C. $=$

9. 5.68×10 ☐ 56.8×100
[Cairo - Al Mostaabal 24]

- A. $>$ B. $<$
C. $=$ D. $\leq$

10. 3.408 ☐ $\frac{348}{100}$

- A. $>$ B. $<$
C. $=$

11. The greatest decimal from the following is _____
[El Monofia - Tala 24]

- A. 0.6 B. 0.06
C. 0.006 D. 0.606

12. $14.1 \text{ } 7 > 14.158$

- A. 3 B. 4
C. 5 D. 6

13. Which of the following is true ?

- A. $0.532 > 0.537$ B. $0.1 + 3 < 1.3$
C. $1.019 > 1.1$ D. $\frac{18}{10} = 1.8$

14. Which of the following is NOT true ?

- A. $14.14 > 14.014$ B. $\frac{143}{100} = 1.43$
C. $2.051 > 2.501$ D. $2.005 < 5.002$

Lesson 5

► Rounding Decimals

Learn Different strategies to round decimals

You can round [approximate] decimal numbers using one of the following strategies :

- 1 Midpoint strategy.
- 2 Rounding rule strategy.

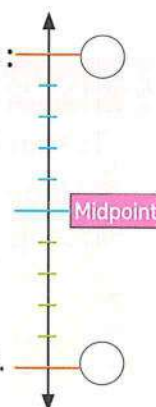
Did You Know?!

Table tennis is one of the world's most popular games. It became an Olympic sport in 1988. A table tennis ball weighs between **2.4** grams and **2.53** grams $\approx$ **2.5** grams

First Midpoint strategy

To round decimals using midpoint strategy, do as follows :

1. Draw a vertical number line.
2. Put the two numbers that the given number lies between them.
3. Put their midpoint.
4. If the given number is at or above the midpoint, round up and if the given number is below the midpoint, round down.



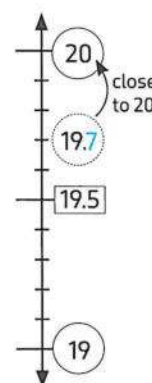
Example 1

Use midpoint strategy to round each of the following.

- a. 19.7 [to the nearest whole number or Unit].
- b. 4.62 [to the nearest Tenth].
- c. 8.765 [to the nearest Hundredth].

Solution

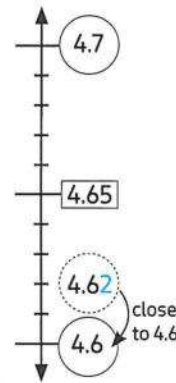
- a. • 19.7 is between 19 and 20
- 19.5 is the midpoint between the two numbers 19 and 20
- 19.7 is closer to 20 because 0.7 is **above** the midpoint, then $19.7 \approx 20$



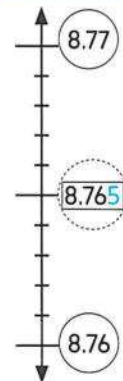
Notes for parents :

- Remind your child with midpoint and rounding rule strategies he/she learned in Primary 4.

- b. • 4.62 is between 4.6 and 4.7
 • 4.65 is the midpoint between the two numbers 4.6 and 4.7
 • 4.62 is closer to 4.6
 because 0.02 is **below** the midpoint
 , then $4.62 \approx 4.6$



- c. • 8.765 is between 8.76 and 8.77
 • 8.765 is the midpoint between the two numbers 8.76 and 8.77
 • 8.765 is closer to 8.77
 because 0.005 is **at** the midpoint
 , then $8.765 \approx 8.77$



Second Rounding rule strategy

To round decimals using rounding rule strategy, do as follows :

1. Underline the digit in the place you want to round the decimal number to it.
2. Look at the digit to its right and circle it.

If

This circled digit is

Less than 5

Leave out the circled digit and the other digits to the right.

Equal to 5 or more

Increase the underlined digit by one, and leave out the other digits to the right.



Example 2

Use rounding rule strategy to round the decimal number 18.5376 to the nearest whole number, Tenth, Hundredth and Thousandth.

Solution

- $18.\underline{5}376 \approx 19$ [to the nearest whole number] (5 = 5)
- $18.5\underline{3}76 \approx 18.5$ [to the nearest Tenth] (3 < 5)
- $18.53\underline{7}6 \approx 18.54$ [to the nearest Hundredth] (7 > 5)
- $18.537\underline{6} \approx 18.538$ [to the nearest Thousandth] (6 > 5)



Remarks

- Rounding to the nearest **Tenth**, the result should include at most **1 decimal** digit
- Rounding to the nearest **Hundredth**, the result should include at most **2 decimal** digits and so on.

Notes for parents :

- Remind your child to round up if the digit to the right of the place value he/she wants to round is equal to or greater than 5

Example 3

Round each number to the place of the underlined digit :

a. 28.12

b. 6.247

c. 12.5928

d. 47.051

e. 5.9184

f. 0.6697

g. 402.601

h. 0.995

Solution

a. $28.\underline{1}2 \approx 28$

b. $6.\underline{2}47 \approx 6.2$

c. $12.59\underline{2}8 \approx 12.593$

d. $47.\underline{0}51 \approx 47.1$

e. $5.9\underline{1}84 \approx 5.92$

f. $0.66\underline{9}7 \approx 0.670$

g. $40\underline{2}.601 \approx 403$

h. $0.9\underline{9}5 \approx 1.00$

Example 4

- Write down the smallest decimal, less than one, that includes only the digits 3, 6, 4 and 2, then round that number to the nearest Hundredth and to the nearest Thousandth.
- Write down the greatest decimal, less than one, that includes 4 digits which are 5, 9, 2 and 7, then round that number to the nearest Hundredth and to the nearest Thousandth.

Solution

- a. To write the smallest decimal less than one, put the decimal point [0.], then write the given digits arranged ascendingly from the left to the right.

- The smallest decimal less than one = 0.2346
- $0.\underline{2}346 \approx 0.23$ [to the nearest Hundredth]
- $0.23\underline{4}6 \approx 0.235$ [to the nearest Thousandth]

- b. To write the greatest decimal, less than one, put the decimal point [0.], then write the given digits arranged descendingly from the left to the right.

- The greatest decimal less than one = 0.9752
- $0.9\underline{7}52 \approx 0.98$ [to the nearest Hundredth]
- $0.97\underline{5}2 \approx 0.975$ [to the nearest Thousandth]



Check your understanding

Round each number to the place of the underlined digit.

a. $8.14\underline{3}7 \approx$ _____

b. $52.\underline{5} \approx$ _____

c. $35.10\underline{7}2 \approx$ _____

d. $17.\underline{9}7 \approx$ _____

e. $55.52\underline{4} \approx$ _____

f. $1.569\underline{8} \approx$ _____

g. $2.435\underline{5} \approx$ _____

h. $\underline{0}.215 \approx$ _____

i. $1.59\underline{5} \approx$ _____

- Remind your child how he/she write the smallest and the greatest decimal formed from given digits.

Exercise

4

on lesson 5

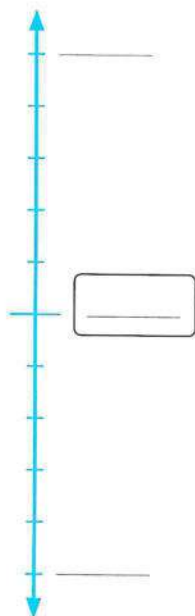
► Rounding Decimals

● REMEMBER ● UNDERSTAND ● APPLY ● PROBLEM SOLVING

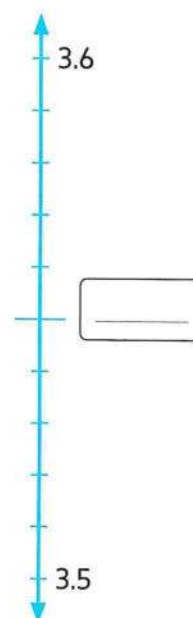
From the school book

1. Label the midpoint of the number line. Place the given decimal number at its proper location.

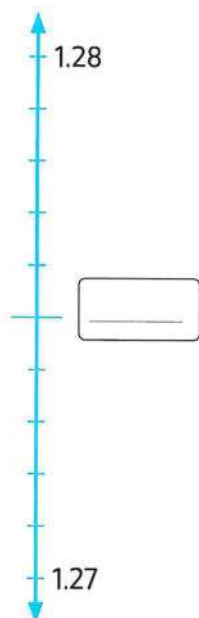
a. Approximate the number 7.7 to the nearest Unit.



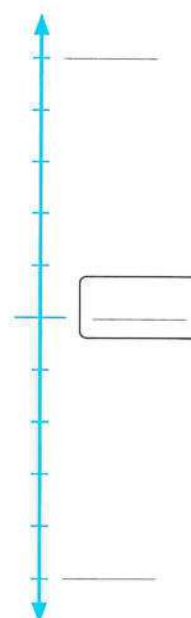
b. Round 3.54 to the nearest Tenth.



c. Round 1.277 to the nearest Hundredth.



d. Round 3.4562 to the nearest Thousandth.



2. Round each of the following numbers to the nearest whole number.

- | | | |
|---|---|---|
| a. $0.9 \approx$ _____
[El Monofia - Ashmoon 24] | b. $0.215 \approx$ _____ | c. $0.512 \approx$ _____ |
| d. $9.9 \approx$ _____ | e. $27.68 \approx$ _____
[Port said - East 24] | f. $18.58 \approx$ _____
[Giza - El Agouza 23] |
| g. $600.601 \approx$ _____ | h. $0.999 \approx$ _____ | i. $0.009 \approx$ _____ |

3. Round each of the following numbers to the nearest Tenth.

- | | | |
|---|--|---------------------------------------|
| a. $8.378 \approx$ _____
[Alexandria 23] | b. $21.729 \approx$ _____
[Qena - Farshut 24] | c. $90.09 \approx$ _____ |
| d. $0.208 \approx$ _____ | e. $73.85 \approx$ _____
[El Monofia - Tala 24] | f. $2.465 \approx$ _____
[Giza 24] |
| g. $476.23 \approx$ _____
[Ismailia 23] | h. $0.07 \approx$ _____ | i. $502\frac{37}{100} \approx$ _____ |

4. Round each of the following numbers to the nearest Hundredth.

- | | | |
|---|--|--|
| a. $1.277 \approx$ _____
[Port Said - Port Fouad 24] | b. $65.567 \approx$ _____
[El Beheira 23] | c. $91.364 \approx$ _____
[El Monofia - Tala 23, Qena 24, Souhag - Tama 24] |
| d. $0.737 \approx$ _____ | e. $0.996 \approx$ _____ | f. $3\frac{8}{1000} \approx$ _____ |

5. Round each of the following numbers to the nearest Thousandth.

- | | | |
|----------------------------|---------------------------|--|
| a. $2.0509 \approx$ _____ | b. $0.0474 \approx$ _____ | c. $4.6798 \approx$ _____ |
| d. $19.9996 \approx$ _____ | e. $0.0004 \approx$ _____ | f. $0.9986 \approx$ _____
[Cairo - El Salam 24] |

6. Round each of the following to the place of the underlined digit.

- | | | |
|---------------------------------------|--------------------------------------|---------------------------------------|
| a. $36.\underline{9}16 \approx$ _____ | b. $5.5\underline{4}8 \approx$ _____ | c. $1.\underline{9}8 \approx$ _____ |
| d. $0.0\underline{8}7 \approx$ _____ | e. $0.\underline{0}81 \approx$ _____ | f. $20.3\underline{6}7 \approx$ _____ |
| g. $3.9\underline{9}8 \approx$ _____ | h. $\underline{0}.09 \approx$ _____ | i. $\underline{0}.8 \approx$ _____ |
| j. $1.\underline{4}99 \approx$ _____ | k. $1.\underline{0}88 \approx$ _____ | l. $1.02\underline{2}9 \approx$ _____ |

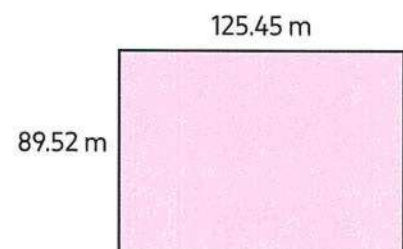
7. Complete the following table as you round each decimal to the stated place value.

	Number	Round to the nearest			
		Whole number [unit]	Tenth	Hundredth	Thousandth
a.	123.3569	123	123.4	123.36	123.357
b.	528.2025	_____	_____	_____	_____
c.	43.5426	_____	_____	_____	_____
d.	21.84792	_____	_____	_____	_____
e.	0.5297	_____	_____	_____	_____
f.	0.0546	_____	_____	_____	_____
g.	4.2688	_____	_____	_____	_____

8. Mazen is planning a trip from Cairo to the waterfall region in Wadi El Rayan. He will travel 147.72 kilometers. Round the distance to the nearest Tenth.

9. Mazen stops to have a snack and stretch after driving 73.255 kilometers. Round the distance to the nearest Hundredth.

10. A farmer is building a new fence for her sheep field. She wants to build a fence around the whole field. Estimate how much fencing you think she will need by rounding each dimension to the nearest Tenth. Explain your thinking.



- 17.**  Read the passage , and then answer the question.

A geologist measured the exact distance between two of the falls at 31.45 meters and between two others at 36.921 meters. If both distances were rounded to the nearest whole number, would they fall into the range given in the passage? Explain your thinking.



18. Complete with suitable digits.

- 49

Multiple Choice Questions

Choose the correct answer.

1. Rounding 0.9 to the nearest whole number is _____.
[Souhag - Gerga 24]
A. 0.9 B. 0
C. 2 D. 1
2. $2.153 \approx$ _____ [to the nearest Tenth]
[Cairo - Hadayek El Quba 24, Giza - Awseem 24]
A. 2.1 B. 2.15
C. 2.14 D. 2.2
3. $21.345 \approx$ _____
[to the nearest Hundredth] [Cairo 23]
A. 21 B. 21.3
C. 21.34 D. 21.35
4. $12.0189 \approx$ _____
[round to nearest Thousandth]
[Alexandria - Agami 24]
A. 12.089 B. 12.018
C. 12.019 D. 10.000
5. $193.2 \approx$ _____ [to the nearest Ten]
[El Monofia - Shebin El Kom 24]
A. 190 B. 200
C. 193 D. 193.4
6. $8.391 \approx 8.4$
[rounded to the nearest _____]
[Port Said - North 24]
A. Hundredth B. Tenth
C. Thousandth D. Tens
7. $82.4973 \approx 82.497$
[to the nearest _____]
A. whole number B. Tenth
C. Hundredth D. Thousandth
8. $9.3 \approx 9$ [to the nearest _____]
[El Menia - Deir Mawas 24]
A. Unit B. Tenth
C. Hundredth D. Thousandth
9. $42.15 \approx$ _____
[to the nearest one decimal place]
[El Monofia - Menof 24, Ashmoon 24, Sers El Laian 24]
A. 42.1 B. 42.2
C. 42.15 D. 42.05
10. $3.649 \approx$ _____
[to the nearest 2 decimal places]
[El Kalyoubia 23]
A. 3.74 B. 3.65
C. 3.54 D. 4.6
11. $2\frac{7}{1,000} \approx$ _____
[to the nearest Hundredth]
A. 2 B. 2.1
C. 2.01 D. 2.007
12. $999.9 \approx$ _____
[to the nearest whole number]
A. 990 B. 999
C. 1,000 D. 900

CONCEPT 2

Adding and Subtracting Decimals

► Lessons 6&7

- Estimating Decimal Sums
- Modeling Decimal Addition

Learning Objectives:

- Students will estimate sums of decimal numbers.
- Students will model decimal addition.

► Lessons 8 to 10

- Modeling Decimal Subtracting
- Estimating Decimal Differences
- Subtracting to the Thousandths Place

Learning Objectives:

- Students will model decimal subtraction.
- Students will estimate differences of decimal numbers.
- Students will apply strategies to subtract decimals to the Thousandths place.
- Students will check the reasonableness of their answers.

► Lesson 11

- Decimal Story Problems

Learning Objectives:

- Students will add and subtract decimal numbers to the Thousandths place to solve story problems.

Fast Fact

The Tahya Misr Bridge in Cairo is **540** meters long and **67.3** meters wide. It holds the world record for the widest cable-stayed bridge in the world.



Lessons 6 & 7

- Estimating Decimal Sums
- Modeling Decimal Addition

Learn 1 Estimating decimal sums

Sameh measured the tallness of his son.
He found that his son is 1.15 meters tall.
Sameh said that his son is about 1 meter tall.

- **Estimation** is a way to get a number that is close to the actual answer but not exact.
- In this lesson, you will learn many ways to estimate decimal sums.



1 Front-end estimation strategy

- Write the first digit of the number from the left as it is.
- Change the rest of digits into zeroes.

For Example :

- 12.18 is closer to 10.00 = 10
- 417.59 is closer to 400.00 = 400

Example 1

Estimate each of the following sums by using front-end estimation.

a. $3.41 + 5.22$

b. $41.925 + 52.236$

Solution 

a. $3.41 + 5.22$

Estimate : $3 + 5 = 8$

b. $41.925 + 52.236$

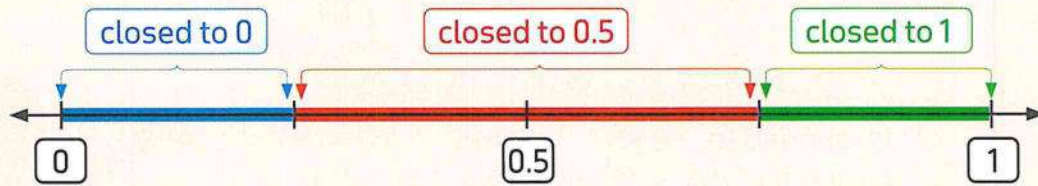
Estimate : $40 + 50 = 90$

Notes for parents :

- Remind your child that he/she just looks at the first digit of the number from the left side, or the highest place value when estimating using front-end strategy.

2 Benchmark decimals strategy

- The benchmark numbers are 0 , $\frac{1}{2}$, 1
- The benchmark decimal for one-half is $0.5 = 0.50 = 0.500$



For Example :

- Each of : 0.1 , 0.01 , 0.001 is closer to 0
- Each of : 0.9 , 0.99 , 0.999 is closer to 1
- Each of : 0.52 , 0.46 , 0.611 , 0.395 is closer to 0.5

Example 2

Estimate each of the following sums by using benchmark decimals.

- | | |
|----------------------|--------------------|
| a. $0.41 + 0.58$ | b. $0.6 + 0.391$ |
| c. $12.492 + 13.659$ | d. $14.999 + 3.01$ |

Solution

a. $0.41 + 0.58$

Estimate : $0.5 + 0.5 = 1$

b. $0.6 + 0.391$

Estimate : $0.5 + 0.5 = 1$

c. $12.492 + 13.659 = 12 + 0.492 + 13 + 0.659$

Estimate : $12 + 0.5 + 13 + 0.5 = 26$

d. $14.999 + 3.01 = 14 + 0.999 + 3 + 0.01$

Estimate : $14 + 1 + 3 + 0 = 18$



Hint

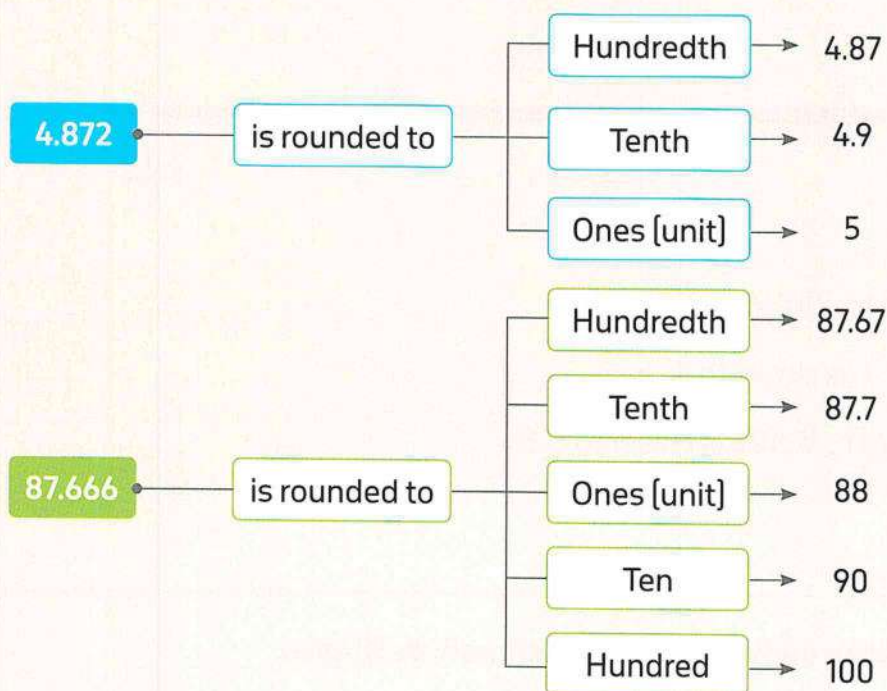
You can separate wholes and parts before using benchmark decimals.

- Remind your child that benchmark decimals are common decimals that he/she can use to judge and compare other decimals.

3 Rounding strategy

You can round decimals in many ways to the nearest Hundredth, Tenth, Ones [unit], Ten, Hundred and so on.

For Example :



Example 3

Estimate the sum $45.561 + 14.047$ by using rounding.

Solution

- $45.561 + 14.047$ Estimate : $50 + 10 = 60$ [to the nearest Ten]
- $45.561 + 14.047$ Estimate : $46 + 14 = 60$ [to the nearest Ones]
- $45.561 + 14.047$ Estimate : $45.6 + 14.0 = 59.6$ [to the nearest Tenth]
- $45.561 + 14.047$ Estimate : $45.56 + 14.05 = 59.61$ [to the nearest Hundredth]

Note that

Rounding to the lowest place value will give you the most accurate estimation.

Check your understanding

Estimate each of the following sums by using more than one strategy.

- $4.39 + 7.12$ _____
- $62.815 + 37.109$ _____
- $15.98 + 24.021$ _____

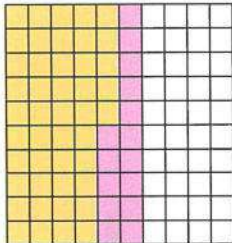
Notes for parents :

- Remind your child to round up if the digit to the right of the place value he/she wants to round is equal to or greater than 5, and round down if it is less than 5.

Learn 2 Modeling decimal addition

To evaluate : $0.45 + 0.15$

- Use two different colors to create a model of the expression : $0.45 + 0.15$



45 Hundredths + 15 Hundredths = 60 Hundredths
So, $0.45 + 0.15 = 0.60$

To add decimal numbers

- Put the decimal points under each other.
- Put zeroes to the right of the last decimal digit, so that each number has the same number of digits after the decimal point.
- Add by starting from the right to the left.

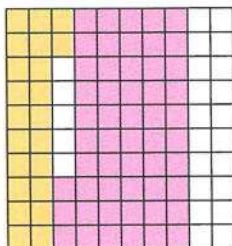
- Use the place-value chart.

Ones				Decimals	
H	T	O	.	Tenths	Hundredths
		0	.	4	5
		0	.	1	5
		0	.	6	0

$$\begin{array}{r}
 \textcircled{1} \\
 0.45 \\
 + 0.15 \\
 \hline
 0.60
 \end{array}$$

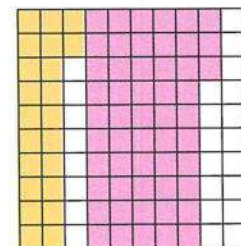
To evaluate : $0.22 + 0.53$

- Use the model.



22 Hundredths + 53 Hundredths = 75 Hundredths
So, $0.22 + 0.53 = 0.75$

Another way of modeling decimal addition :



- Use the place-value chart.

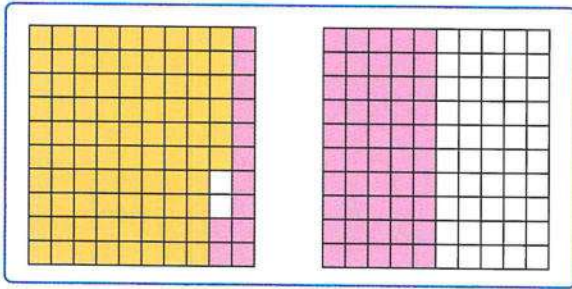
Ones				Decimals	
H	T	O	.	Tenths	Hundredths
		0	.	2	2
		0	.	5	3
		0	.	7	5

$$\begin{array}{r}
 0.22 \\
 + 0.53 \\
 \hline
 0.75
 \end{array}$$

- Make sure that when your child adds decimals, he/she puts the decimal points under each other.

To evaluate : $0.86 + 0.62$

- Use the model.



86 Hundredths
+ 62 Hundredths
= 148 Hundredths
So, $0.86 + 0.62 = 1.48$

- Use the place-value chart.

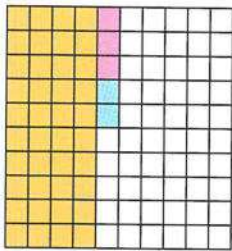
Ones				Decimals	
H	T	O	.	Tenths	Hundredths
		0	.	8	6
		0	.	6	2
1			.	4	8

$$\begin{array}{r}
 \textcircled{1} \\
 0.86 \\
 + 0.62 \\
 \hline
 1.48
 \end{array}$$



To evaluate : $0.4 + 0.03 + 0.02$

- Use the model.



4 Tenths + 3 Hundredths + 2 Hundredths
= 40 Hundredths + 3 Hundredths + 2 Hundredths
= 45 Hundredths
So, $0.4 + 0.03 + 0.02 = 0.45$

- Use the place-value chart.

Ones				Decimals	
H	T	O	.	Tenths	Hundredths
		0	.	4	0
		0	.	0	3
		0	.	0	2
0			.	4	5

$$\begin{array}{r}
 0.40 \\
 + 0.03 \\
 + 0.02 \\
 \hline
 0.45
 \end{array}$$



Notes for parents :

- Remind your child that there are more than one model for any addition statement.

To evaluate : $2,923.42 + 4,581.3$

It is impossible to use the model

So, use the place-value chart.

Thousands	Ones				Decimals	
O	H	T	O	.	Tenths	Hundredths
2	9	2	3	.	4	2
4	5	8	1	.	3	0
7	5	0	4	.	7	2

$$\begin{array}{r}
 \textcircled{1} \quad \textcircled{1} \\
 2,923.42 \\
 + 4,581.30 \\
 \hline
 7,504.72
 \end{array}$$



Example 4

Add each of the following.

a. $3.13 + 5.49$

b. $127.3 + 16.45$

c. $14.72 + 7.5 + 0.231$

Solution

a.
$$\begin{array}{r}
 \textcircled{1} \\
 3.13 \\
 + 5.49 \\
 \hline
 8.62
 \end{array}$$

b.
$$\begin{array}{r}
 \textcircled{1} \\
 127.30 \\
 + 16.45 \\
 \hline
 143.75
 \end{array}$$

c.
$$\begin{array}{r}
 \textcircled{1} \quad \textcircled{1} \\
 14.720 \\
 + 7.500 \\
 + 0.231 \\
 \hline
 22.451
 \end{array}$$

Note

You can add decimals horizontally as follows :

$$\begin{array}{c}
 \textcircled{1} \\
 3.13 + 5.49 = 8.62
 \end{array}$$

Check your understanding

Add the following.

a. $0.21 + 0.575$

b. $213.01 + 27.992$

c. 3 Hundredths + 4 Thousandths = _____

d. 13 Hundredths + 65 Thousandths = _____

- Let your child learn that the modeling decimal adding strategy is impossible to use when adding large numbers.

Exercise

5

on lessons 6&7

- Estimating Decimal Sums
- Modeling Decimal Addition

REMEMBER

UNDERSTAND

APPLY

PROBLEM SOLVING

From the school book

1. Estimate each of the following sums.

a. $0.52 + 0.49$

Estimate _____

c. $7.99 + 4.011$

Estimate _____

e. $42.998 + 42.091$

Estimate _____

b. $3.451 + 8.091$

Estimate _____

d. $9.98 + 4.56$

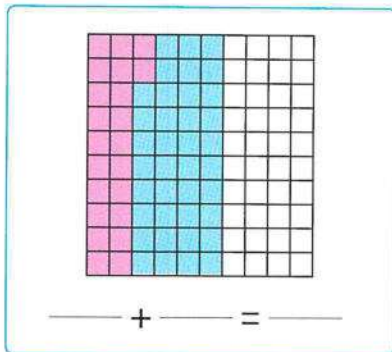
Estimate _____

f. $4.981 + 5.019$

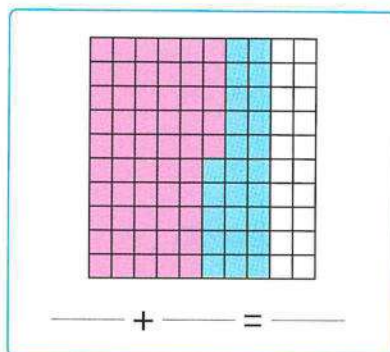
Estimate _____

2. Write an expression to match each of the following models, then use each model to evaluate the expression.

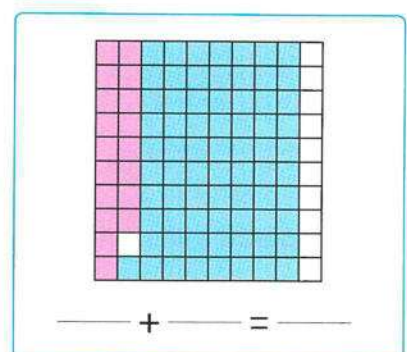
a.



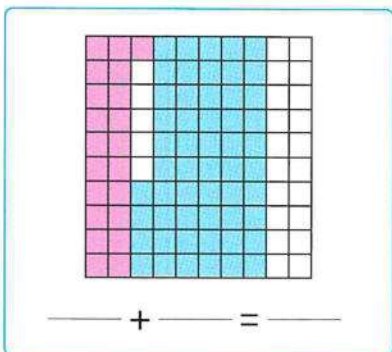
b.



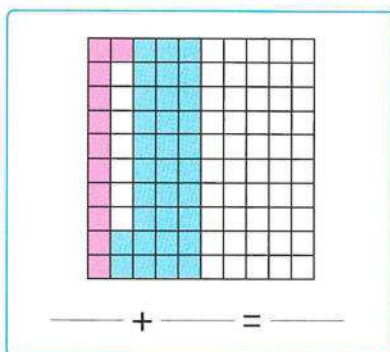
c.



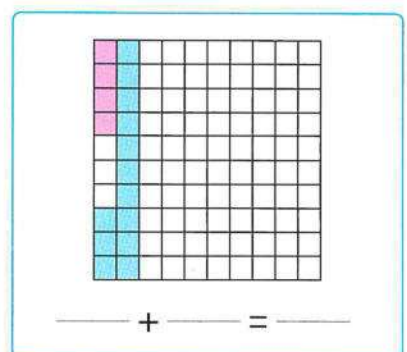
d.



e.



f.



3. Complete each of the following.

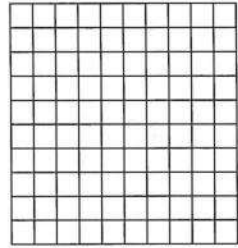
a.  Estimate $0.13 + 0.23$ _____

- Use two different colors to create a model of the expression $0.13 + 0.23$

- Record 0.13 and 0.23 in the place-value chart.

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths

- Evaluate : $0.13 + 0.23 =$ _____

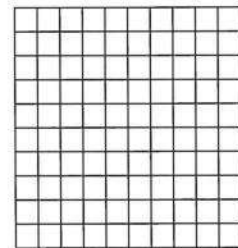
b.  Estimate $0.05 + 0.05$ _____

- Use two different colors to create a model of the expression $0.05 + 0.05$

- Record 0.05 and 0.05 in the place-value chart.

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths

- Evaluate : $0.05 + 0.05 =$ _____

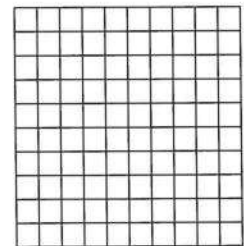
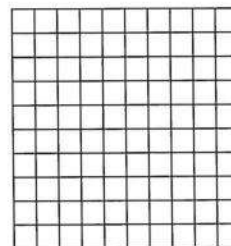
c.  Estimate $0.45 + 0.84$ _____

- Use two different colors to create a model of the expression $0.45 + 0.84$

- Record 0.45 and 0.84 in the place-value chart.

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths

- Evaluate : $0.45 + 0.84 =$ _____

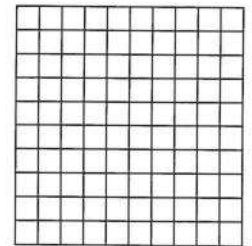
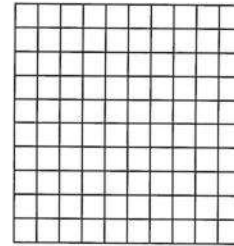


d.  Estimate $0.92 + 0.89$ _____

- Use two different colors to create a model of the expression $0.92 + 0.89$

- Record 0.92 and 0.89 in the place-value chart.

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths



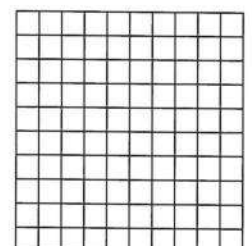
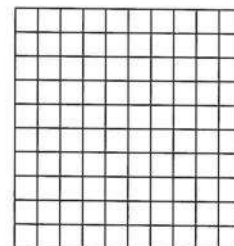
- Evaluate : $0.92 + 0.89 =$ _____

e.  Estimate $0.97 + 0.42$ _____

- Use two different colors to create a model of the expression $0.97 + 0.42$

- Record 0.97 and 0.42 in the place-value chart.

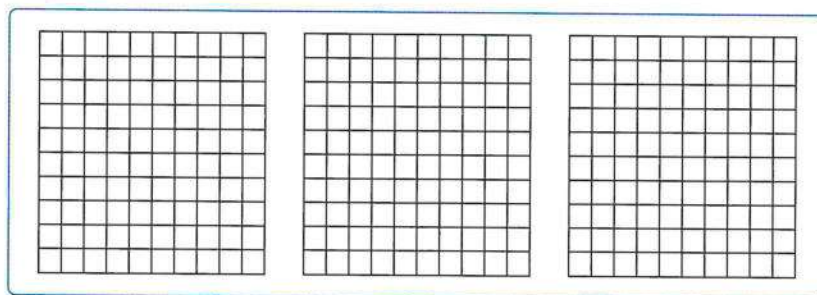
Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths



- Evaluate : $0.97 + 0.42 =$ _____

f. Estimate $1.9 + 0.62$ _____

- Use two different colors to create a model of the expression $1.9 + 0.62$



- Record 1.9 and 0.62 in the place-value chart.

Thousands	Ones			.	Decimals	
O	H	T	O	.	Tenths	Hundredths

- Evaluate : $1.9 + 0.62 =$ _____

4. Find the result of each of the following.

$$\begin{array}{r} \text{a.} \quad 0.231 \\ + 0.754 \\ \hline \end{array}$$

$$\begin{array}{r} \text{b.} \quad 2.53 \\ + 0.19 \\ \hline \end{array}$$

$$\begin{array}{r} \text{c.} \quad 4.89 \\ + 0.87 \\ \hline \end{array}$$

$$\begin{array}{r} \text{d.} \quad 16.34 \\ + 8.79 \\ \hline \end{array}$$

$$\begin{array}{r} \text{e.} \quad 7.51 \\ + 6.492 \\ \hline \end{array}$$

$$\begin{array}{r} \text{f.} \quad 967.63 \\ + 91.2 \\ \hline \end{array}$$

$$\begin{array}{r} \text{g.} \quad 48.42 \\ + 59.096 \\ \hline \end{array}$$

$$\begin{array}{r} \text{h.} \quad 35.001 \\ + 14.999 \\ \hline \end{array}$$

$$\begin{array}{r} \text{i.} \quad 4.15 \\ + 8.6 \\ + 9.283 \\ \hline \end{array}$$

5. Find the result of each of the following.

$$\text{a. } 35.46 + 123.32 = \underline{\hspace{2cm}} \quad \text{[Giza - South 24]}$$

$$\text{b. } 4.8 + 4.08 = \underline{\hspace{2cm}} \quad \text{[Port Said 24]}$$

$$\text{c. } 45.321 + 23.65 = \underline{\hspace{2cm}} \quad \text{[Kafra El Sheikh 24]}$$

$$\text{d. } 12.06 + 14.9 = \underline{\hspace{2cm}} \quad \text{[Souhag - Akhmeem 24]}$$

$$\text{e. } 62.6 + 8.214 = \underline{\hspace{2cm}} \quad \text{[El Menia - Bani Mazar 24]}$$

$$\text{f. } 45.9 + 13.33 = \underline{\hspace{2cm}} \quad \text{[Alexandria - Montaza 24]}$$

$$\text{g. } 38.73 + 24.5 = \underline{\hspace{2cm}} \quad \text{[El Monofia - El Bagour 24]}$$

$$\text{h. } 24.66 + 17.34 = \underline{\hspace{2cm}} \quad \text{[Aswan 24]}$$

$$\text{i. } 1.007 + 9 = \underline{\hspace{2cm}}$$

$$\text{j. } 13 + 2.75 = \underline{\hspace{2cm}}$$

$$\text{k. } 213.01 + 27.99 = \underline{\hspace{2cm}}$$

$$\text{l. } 12.179 + 11\frac{1}{4} = \underline{\hspace{2cm}}$$

6. Find the result of each of the following.

$$\text{a. } 37.42 + 43.01 + 19.15 = \underline{\hspace{2cm}}$$

$$\text{b. } 28.65 + 17.3 + 2.05 = \underline{\hspace{2cm}}$$

$$\text{c. } 6 + 3.65 + 4.912 = \underline{\hspace{2cm}}$$

$$\text{d. } 53.245 + 1.97 + 213.8 = \underline{\hspace{2cm}}$$

7. Complete the missing digits.

$$\begin{array}{r} \text{a.} \quad 3.\square7 \\ + 6.5\square \\ \hline \square.89 \end{array}$$

$$\begin{array}{r} \text{b.} \quad 1.\square6 \\ + 3.3\square \\ \hline \square.33 \end{array}$$

$$\begin{array}{r} \text{c.} \quad 97.48 \\ + 43.\square\square \\ \hline \square\square\square.93 \end{array}$$

8. Complete the following.

- a. 4 Thousandths + 3 Thousandths = _____ Thousandths. [El Menia - Mallawi 24]
- b. The sum of $3.127 + 8.65 =$ _____. [Luxor 24]
- c. The sum of $2.817 + 1.183 =$ _____. [Giza - Awseem 24]
- d. 9 Hundredths + 56 Hundredths = _____ Hundredths. [Cairo - Heliopolis 23]
- e. 2 Thousandths + 3 Hundredths = _____ Thousandths. [Cairo - El Sherouk 23]
- f. 8 Thousandths + 95 Hundredths = _____ Thousandths. [Ismailia 24]
- g. 5 Tenths + 5 Thousandths = _____ Thousandths. [Cairo - El Marg 23]

9. Noha saved 18.57 pounds and her sister saved 19.3 pounds. Find the sum they saved ?

[Cairo - El Sayeda Zeinab 24]

10. Bassem bought two watermelons, the mass of the first is 2.62 kg and the mass of the second is 2.71 kg. What is the sum of their masses together ?

[Alexandria - El Gamarek 24]

11. If Mona's mass is 55.45 kg. Then if her mass increased after a month by 3.15 kg , what is her mass ?

[El Menia - Samalout 24]

12. 📖 If a farmer can lift 94.635 liters of water a minute in his shadoof, about how many liters can he lift in 4 minutes ?

13. 📖 Samar wanted to ride her bike 40 kilometers this week. By Thursday , she had ridden 34.99 kilometers. On Friday, she rode 4.01 kilometers. Estimate to see if she has met her goal.

Estimate : _____

14. 📖 Taha has 54.20 L.E. His brother has 45.75 L.E. They want to combine their money to purchase a box of apples for 100 L.E. Estimate to see if they have enough money.

Estimate : _____



Multiple Choice Questions

Choose the correct answer.

1. The benchmark of 0.99 is _____
[Cairo - El Sahel 24]

A. 0 B. 0.5
C. 1 D. 1.5

2. The estimate of the sum of
 $35.762 + 63.014$ is _____
[Cairo - Al Khalifa and Al Mokattam 23]

A. 99 B. 80
C. 98.76 D. 110

3. $4.6 + 0.06 =$ _____
[Cairo - Ain Shams 24]

A. 4.66 B. 4.6
C. 5.2 D. 4.12

4. $0.05 + 0.05 =$ _____ [Port Said 23]
A. 0.55 B. 0.1
C. 10 D. 5.5

5. $2 + 0.05$ $1.7 + 0.7$
[El Menia - Mallawi 24]
A. < B. = C. >

6. $1.7 + 0.2$ $1.33 + 0.51$
A. < B. = C. >

7. $20 + 0.078 =$ _____
[Cairo - El Sherouk 23]
A. 20.078 B. 20.78
C. 20.708 D. 20.807

8. $16.9 + 2.185 =$ _____
[Alexandria - Agmi 24]
A. 19.085 B. 18.194
C. 18.085 D. 17.084

9. 7 Tenths + 3 Tenths = _____
[Cairo - El Salam 23]
A. 1 B. 10
C. 100 D. 1,000

10. 4 Thousandths + 3 Thousandths
= _____ Thousandths. [Port Said 23]
A. 7,000 B. 7
C. 0.7 D. 0.07

11. 4 Hundredths + 35 Thousandths
= _____
A. 0.39 B. 0.039
C. 0.07 D. 0.075

12. 3 Hundredths + 5 Tenths
= _____ Hundredths.
[Kaf El Sheikh - Bayala 24]
A. 8 B. 35
C. 53 D. 3

Lessons 8 to 10

- Modeling Decimal Subtracting
- Estimating Decimal Differences
- Subtracting to the Thousandths Place

Learn 1 Estimating decimal differences

You can use the strategies of estimation that you studied in the previous lesson to estimate decimal differences as the following example.



Example 1

Estimate each of the following.

a. $39.79 - 20.027$

b. $18.95 - 11.7$

c. $0.88 - 0.72$

Solution

a. $39.79 - 20.027$

estimate : $40 - 20 = 20$

[if you round to the nearest Ten]

b. $18.95 - 11.7$

estimate : $19 - 12 = 7$

[if you round to the nearest Ones]

c. $0.88 - 0.72$

estimate : $0.9 - 0.7 = 0.2$

[if you round to the nearest Tenth]

$0.88 - 0.72$

estimate : $1 - 1 = 0$

[if you round to the nearest Ones]



check your understanding

Estimate each of the following.

a. $0.92 - 0.76$

b. $17.01 - 13.9$

c. $140.61 - 99.43$

Notes for parents :

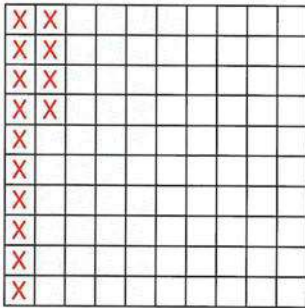
- Remind your child that estimation is a way to get a number that is close to another number but not exact.

Learn 2 Modeling Decimal Subtracting

To evaluate : $0.52 - 0.14$

1. Shade a model to represent the minuend [0.52].
2. Add X's to represent the subtrahend [0.14].
3. Count the shaded squares without [X] which is the difference.

• Use the model.



• Use the place-value chart.

Ones				Decimals	
H	T	O	.	Tenths	Hundredths
		0	.	5	2
		0	.	1	4
		0	.	3	8

$$\begin{array}{r} \overset{4}{5} \overset{12}{2} \\ - 0. \\ \hline 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \end{array}$$

To subtract decimal numbers

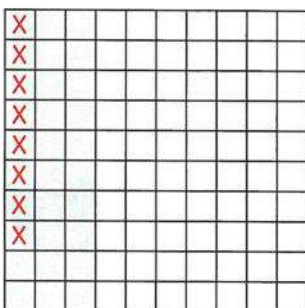
- 1 Put the decimal points under each other.
- 2 Put zeroes to the right of the last decimal digit, so that each number has the same number of digits after the decimal point.
- 3 Subtract by starting from the right to the left.

- You can subtract the previous numbers horizontally as follows :

$$\overset{4}{0.} \overset{12}{\cancel{5}2} - \overset{1}{0.} \overset{4}{\cancel{1}4} = 0.\overset{3}{\cancel{5}2}$$

To evaluate : $0.3 - 0.08$

• Use the model.



• Use the place-value chart.

Ones				Decimals	
H	T	O	.	Tenths	Hundredths
		0	.	3	0
		0	.	0	8

$$\begin{array}{r} \overset{2}{3} \overset{10}{0} \\ - 0. \\ \hline 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \\ 0. \end{array}$$

Note that

Adding zeroes to the right of the last decimal digit does not change its value.

- Remind your child to put the decimal points under each other when subtracting decimals.

To evaluate : $3,204.4 - 1,823.015$

- It is impossible to use the model. **So**, use the place-value chart.

Thousands	Ones			.	Decimals		
	H	T	O	.	Tenths	Hundredths	Thousandths
0	2	0	4	.	4	0	0
1	8	2	3	.	0	1	5
1	3	8	1	.	3	8	5

$$\begin{array}{r} 11 \\ 2 10 3 9 10 \\ \cancel{3} \cancel{2} \cancel{0} 4 \cancel{4} \cancel{0} \cancel{0} \\ - 1 8 2 3 0 1 5 \\ \hline 1 3 8 1 3 8 5 \end{array}$$

Example 2

Subtract each of the following.

a. $5.43 - 2.21$

b. $8.6 - 6.51$

c. $20 - 11.624$

Solution

a.

$$\begin{array}{r} 5.43 \\ - 2.21 \\ \hline 3.22 \end{array}$$

b.

$$\begin{array}{r} 8. \cancel{6} \cancel{0} \\ - 6. 5 1 \\ \hline 2. 0 9 \end{array}$$

c.

$$\begin{array}{r} 199910 \\ \cancel{20000} \\ - 11624 \\ \hline 8376 \end{array}$$



check your understanding

Subtract each of the following.

a. $2.325 - 0.214$

b. $12.78 - 3.5$

c. 97 Thousandths – 49 Thousandths.

d. 7 Hundredths – 32 Thousandths.

Notes for parents :

- Remind your child that adding zeroes to the right of the last decimal digit does not change its value.

Exercise

6

on lessons 8 to 10

- Modeling Decimal Subtracting
- Estimating Decimal Differences
- Subtracting to the Thousandths Place

REMEMBER

UNDERSTAND

APPLY

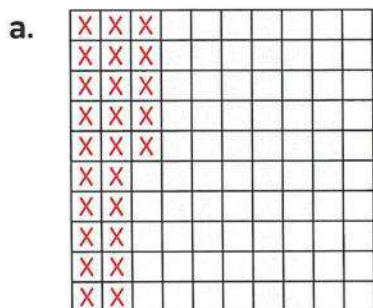
PROBLEM SOLVING

From the school book

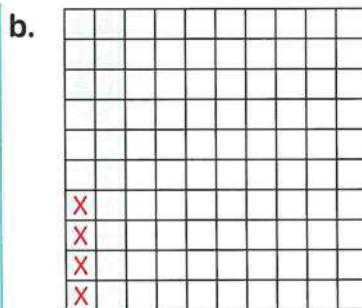
1. Estimate each of the following.

- $2.62 - 1.59$ estimate _____
-  $2.419 - 1.240$ estimate _____
-  $35.9 - 10.8$ estimate _____
- $214.024 - 113.78$ estimate _____
- $0.951 - 0.729$ estimate _____

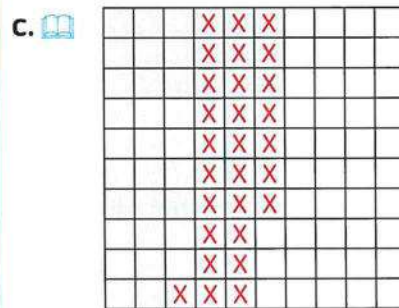
2. Write an expression to match each of the following models, then use each model to evaluate the expression.



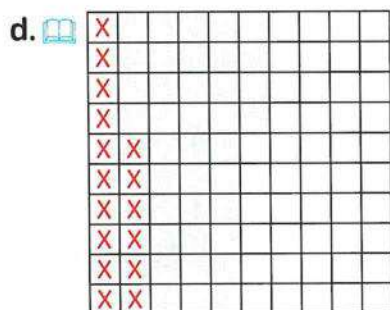
_____ - _____ = _____



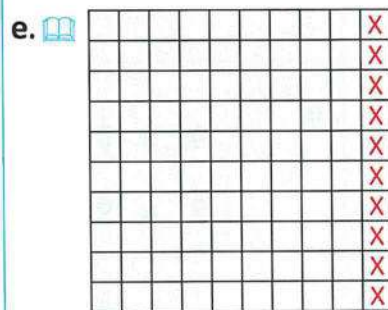
_____ - _____ = _____



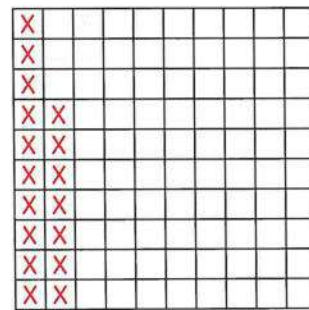
_____ - _____ = _____



_____ - _____ = _____

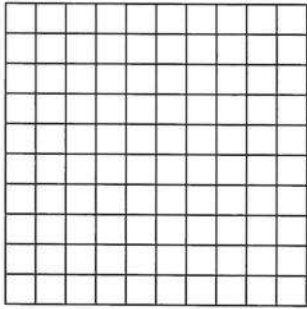


_____ - _____ = _____

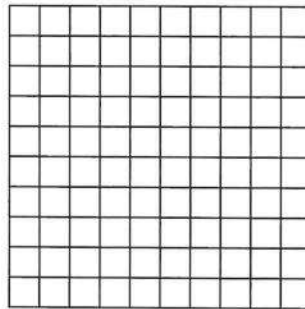


3. Create a model to match each of the following expressions and evaluate each of them.

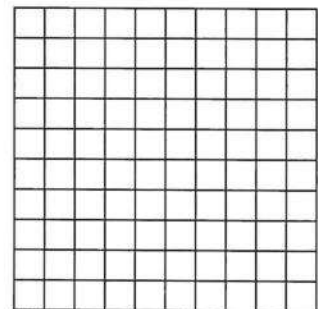
a. $0.67 - 0.49 =$ _____



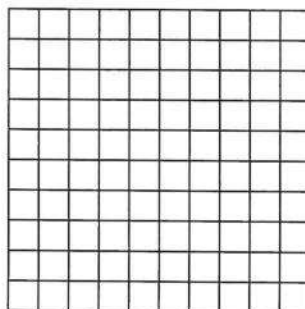
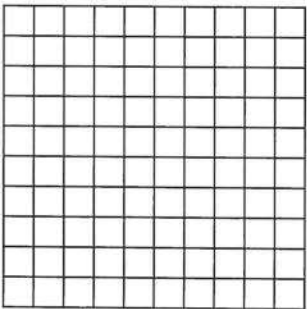
b. $0.1 - 0.09 =$ _____



c. $0.39 - 0.13 =$ _____



d. $1.23 - 1.02 =$ _____



4. Find the result of each of the following.

a.

$$\begin{array}{r} 0.781 \\ - 0.531 \\ \hline \end{array}$$

b.

$$\begin{array}{r} 0.593 \\ - 0.194 \\ \hline \end{array}$$

c.

$$\begin{array}{r} 0.5 \\ - 0.375 \\ \hline \end{array}$$

d.

$$\begin{array}{r} 3,218.975 \\ - 218.853 \\ \hline \end{array}$$

e.

$$\begin{array}{r} 4,524.62 \\ - 2,498.124 \\ \hline \end{array}$$

f.

$$\begin{array}{r} 4,611.3 \\ - 1,982.45 \\ \hline \end{array}$$

5. Find the result of each of the following.

a. $4.185 - 3.4 =$ _____

[El Menia - Matai 24]

b. $4.66 - 2.09 =$ _____

[Aswan - Kom Ombo 23]

c. $8.659 - 4.32 =$ _____

[Cairo - El Marg 23]

d. $63 - 12.3 =$ _____

[Cairo - Rod El Farg 24]

e. $0.9 - 0.889 =$ _____

f. $12.74 - 0.359 =$ _____

g. $5.27 + 8.39 - 3\frac{14}{100} =$ _____

h. $512 + 88.35 - 67.035 =$ _____

6. Complete the table.

The expression	Estimating difference	Actual difference
a. $4.45 - 4.32 =$	_____	_____
b. $0.97 - 0.82 =$	_____	_____
c. $5.05 - 4.15 =$	_____	_____
d. $29.98 - 11.99 =$	_____	_____

7. Complete.

a. $1.6 +$ _____ $= 9.6$

b. _____ $+ 3.9 = 6.5$

c. _____ $+ 54.8 = 77.59$

d. $85.47 +$ _____ $= 100$

e. $6.27 -$ _____ $= 3.286$

f. _____ $- 3\frac{3}{5} = 7.634$

g. $33.3 -$ _____ $= 12.008$

h. _____ $- 41.41 = 3.8$

[Cairo - Al Mokattam 24]

8. Find the missing digits.

a.
$$\begin{array}{r} 5.69 \\ - 4.\square\square \\ \hline \square.45 \end{array}$$

b.
$$\begin{array}{r} 9.51\square \\ - \square.\square\square1 \\ \hline 4.242 \end{array}$$

c.
$$\begin{array}{r} 113.57\square \\ - 13.\square\square8 \\ \hline \square\square\square.17\square \end{array}$$

d.
$$\begin{array}{r} 5.\square8\square \\ - 1.413 \\ \hline \square.3\square7 \end{array}$$

e.
$$\begin{array}{r} 299.\square\square\square \\ - \square\square.457 \\ \hline 243.\square3\square \end{array}$$

f.
$$\begin{array}{r} \square\square.\square\square \\ - 23.97 \\ \hline 18.95 \end{array}$$

9. Put the suitable relation ($<$, $=$ or $>$).

a. $3.5 - 2.1$ $3.5 + 2.1$

[Alexandria - First Montaza 23]

c. $7.32 - 1.93$ $6.78 - 0.42$

e. $99.89 - 90.09$ $10 - 1.01$

g. $6.18 + 3.82$ $87.56 - 77.5$

b. $1.471 - 0.53$ 0.951

d. $58.003 - 57.03$ $1 + 0.973$

f. $4.722 - 0.009$ $8 - 3.228$

h. $0.2 - 0.05$ $4.9 - 4.75$

10. Evaluate each difference. Then identify each digit's place value.

a. 98 Thousandths $-$ 5 Thousandths $=$ _____ Thousandths

Place value : _____ Hundredths and _____ Thousandths

b.  57 Thousandths $-$ 12 Thousandths $=$ _____ Thousandths

Place value : _____ Hundredths and _____ Thousandths

c.  32 Thousandths $-$ 15 Thousandths $=$ _____ Thousandths

Place value : _____ Hundredths and _____ Thousandths

d.  5 Hundredths $-$ 24 Thousandths $=$ _____ Thousandths

[Assuit 24]

Place value : _____ Hundredths and _____ Thousandths

e. 7 Hundredths $-$ 17 Thousandths $=$ _____ Thousandths

Place value : _____ Hundredths and _____ Thousandths

f. 8 Tenths $-$ 42 Thousandths $=$ _____ Thousandths

Place value : _____ Tenths, _____ Hundredths and _____ Thousandths



Multiple Choice Questions

Choose the correct answer.

1. $2.419 - 1.240 =$ _____

[Aswan 23]

- A. 1.230 B. 1.179
C. 1.239 D. 3.659

2. Estimate $56.25 - 20.98$ is _____

[Cairo - El Sahel 24]

- A. 31 B. 36
C. 35 D. 25

3. 7 Tenths - 7 Hundredths = _____

[Cairo - El Basateen and El Salam 24]

- A. 0 B. 0.63
C. 0.693 D. 0.963

4. 7 Tenths - 63 Hundredths

= _____ Hundredths. [Cairo - El Nouzha 23]

- A. 70 B. 700
C. 7 D. 7.000

5. 7 Hundredths - 7 Thousandths
= _____ Thousandths.

[Cairo - Al Khalifa and Al Mokattam 23]

- A. 7 B. 0
C. 63 D. 77

6. $55.5 - 5.55 =$ _____

[Ismailia 24]

- A. 49.59 B. 49.95
C. 50.5 D. 5.05

7. $4.45 - 4.32$ ○ $1.01 + 0.3$ [Assiut 24]

- A. > B. <
C. =

8. $3.2 + 4.05$ ○ $7.05 + \frac{1}{2}$ [Giza - Awseem 23]

- A. > B. <
C. =

9. $94.8 - 9.82 = 84.46$

- A. 1 B. 2
C. 3 D. 4

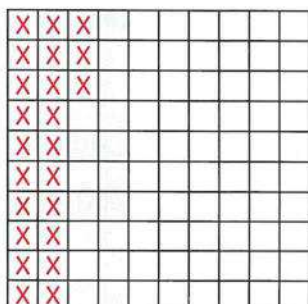
10. $15 - 3.5 =$ _____

[Aswan - Edfu 24]

- A. 11.5 B. 11
C. 12.5 D. 12

11. Which of the following expressions represents the model?

- A. $0.23 - 0.04$
B. $0.4 - 0.23$
C. $0.04 - 0.023$
D. $40 - 23$



12. $13.58 -$ _____ $= 9.89$

[El Monofia - Quessna 24]

- A. 3.69
B. 4.31
C. 38.85
D. 30.69

► Decimal Story Problems

Learn How to solve story problems?

-  1. Read carefully and determine what is being asked.
-  2. Plan and write an equation or expression to solve the story problem.
-  3. Solve the problem and be sure to include units in your answer.
-  4. Check the reasonableness of your answers.

 **Read to understand**
 **Plan**
 **Solve**
 **Check**
Example 1

Soha saved 17.25 L.E. and her brother Amgad saved 8.5 L.E.

Find the sum they saved.

Solution 

The sum they saved = $17.25 + 8.5 = 25.75$ L.E.

**Example 2**

Wael has 14.75 pounds and his sister Mariam has 950 piasters.

Find the difference between what they have in pounds.

Solution 

The difference = $14.75 \text{ pounds} - 950 \text{ piasters}$
 $= 14.75 \text{ pounds} - 9.5 \text{ pounds} = 5.25 \text{ pounds}$

**Example 3**

Waleed bought a pair of trousers for 89.6 L.E. and a shirt for 30.75 L.E., if he gave 200 L.E. to the shopkeeper,

how much change remained with Waleed ?

Solution 



- The price of pair of trousers and shirt = $89.6 + 30.75 = 120.35$ L.E.
- The change remained with Waleed = $200 - 120.35 = 79.65$ L.E.

Notes for parents :

- Some story problems have hidden question or questions that must be answered before you can solve the problem. You have to determine what operation to use and what strategies you will use to help you figure out how to solve the problem.

Exercise 7

on lesson 11

► Decimal Story Problems

● REMEMBER ● UNDERSTAND ● APPLY ● PROBLEM SOLVING

From the school book

1. Ola saved 23.75 pounds and her sister saved 57.34 pounds.
Find the total with them. [El Fayoum 24]



2. Salma has 90.5 pounds, she bought a toy by 64.75 pounds
How much money is remaining with Salma ? [El Kalyoubia 23]



3. Mona had 78.4 L.E. She spent 52.74 L.E.
Find the remainder with her. [El Beheira 24]



4. A merchant has 38.5 meters of cloth and sold 15.5 meters
of it. [Aswan - Edfo 24]
How much does he have left ?



5. Fares bought 9.8 kilograms of apples, 4.6 kilograms of fig.
Find the total weight of apple and fig together ?

[Cairo - El Zaiton 23]



6. The fuel tank in the car was filled with 35 liters of
gasoline, and at the end of the day 15.5 liters of fuel
remained in the tank.
How much fuel did the car consume for that day in liters ?

[El Monofia - Menof 24, Sers El Laian 24]



7. Hatem climbed 5.6 m and Nagy climbed 2.9 m.

What is difference between them ?

[El Monofia - Shebin El Kom 24]



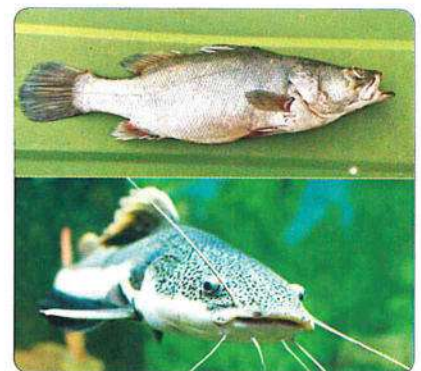
8. Hanaa has 200 pounds. She wants to buy a pair of shoes for 99.8 L.E. , a bag for 45.75 L.E. and a dress for 70.25 L.E.

Can she buy all what she wants ? Why ?



9. Nile perch is 110 centimeters long and more than 5 years old. It weighs 113.39 kilograms and the vundu catfish weighs 38.1 kilograms and is 188 centimeters long.

What is the total mass of both the Nile perch and the vundu catfish ?



10. Read the passage and then respond to the questions.

You will now travel from Khartoum to Juba in South Sudan to see the source of the White Nile. This trip is 1,941.2 kilometers. Juba is also on the bank of the White Nile.

From Juba, you will travel on to Jinja, Uganda. It is a distance of 687.9 kilometers.

Jinja is located near the source of the White Nile.

How long is your journey from Khartoum to Jinja ?

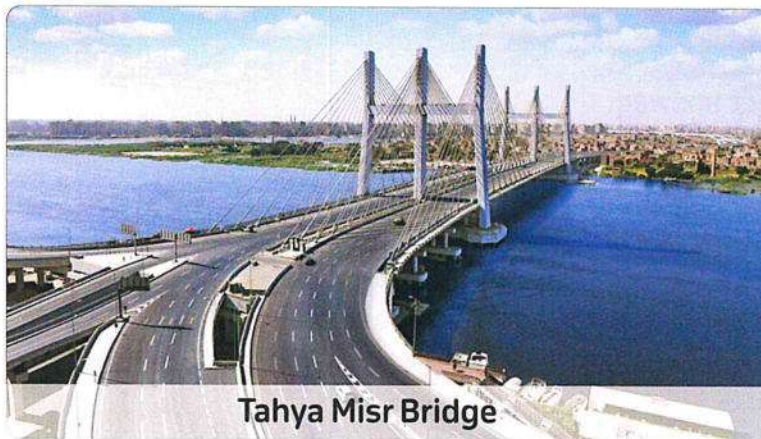
- a. Copy the place-value chart and record the addends.

Thousands	Ones			.	Decimals		
O	H	T	O	.	Tenths	Hundredths	Thousandths
				.			
				.			

- b. Write and solve an addition equation using the two decimal numbers.

_____ + _____ = _____

11.  Read the passage and answer the questions.

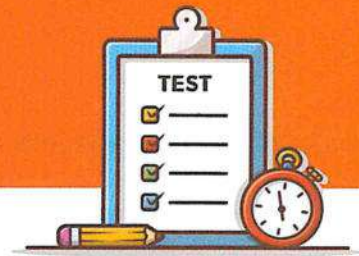


Tahya Misr Bridge

The Tahya Misr Bridge was built in 2016 in Cairo. It serves as a connector across the Nile from northern and eastern Cairo to western Cairo. The bridge is 540 meters long and 67.3 meters wide. It holds the world record for the widest cable-stayed bridge in the world. The longest cable-stayed bridge is the Jiaying-Shaoxing Sea Bridge in China. It is 11.7 meters thinner than the Tahya Misr Bridge. How wide is the Jiaying-Shaoxing Sea Bridge ?

12.  The total length of the Tahya Misr Bridge is 16.7 kilometers. If Rami travels the length of the Tahya Misr Bridge and then returns, how many kilometers in total did he travel ? Write an equation and your answer.
13.  The total length of the Tahya Misr Bridge is 16.7 kilometers. Salem rode his bike along the pedestrian section of the bridge. He rode 3.25 kilometers before he had a flat tire. How many more kilometers does he need to travel ?
14.  The Tahya Misr Bridge was built using 200 cranes. The cranes varied in size and weighed between 6.44 and 544.3 tons [1 ton = 1,000 kilograms]. What is the difference between the lightest crane and the heaviest crane ?
15.  Rashad and his father went on a fishing trip to Lake Nassar. They each caught a huge vundu catfish. The first one weighed 53.25 kilograms and the smaller one weighed 46.8 kilograms. How much did the fish weigh in all ?

Unit One Assessment



1. Choose the correct answer.

1. The value of the digit 7 in the number 5.167 is _____
[Cairo - El Salam 24, Port Said 24, Giza - El Haram 24]
A. 0.7 B. 0.07 C. 700 D. 0.007
2. Rounding the number 56.284 to the nearest Hundredth is _____
[Aswan - Kom Ombo 23]
A. 56.28 B. 56.82 C. 56.3 D. 56.29
3. $9.4 - 5.03 =$ _____
[Kafr El Sheikh - Bayala 24, Souhag 24]
A. 4.37 B. 43.7 C. 4.43 D. 4.1
4. $10 \times 150 =$ _____ Hundreds
[Giza - South 24]
A. 15 B. 150 C. 1,500 D. 1.5
5. A car covers 2.5 km in one minute, then the distance covered in 3 minutes = _____ km
[Cairo - El Salam 23]
A. 7.5 B. 5.7 C. 7 D. 5.4
6. $3.6 + 5.411 =$ _____
[Ismailia - El Kasasen 24]
A. 5.447 B. 8.1011 C. 8.417 D. 9.011
7. The benchmark number of the decimal fraction 0.8 is _____
[Cairo - Helwan 24]
A. zero B. 0.25 C. 0.5 D. 1

2. Complete the following.

1. $9.865 \approx$ _____ [rounding to the nearest Tenth] [Giza 24]
2. $8 + 0.2 + 0.03 + 0.006 =$ _____ [in standard form] [El Beheira 23]
3. $39.543 \approx$ _____ [to the nearest one decimal place] [Cairo - El Maadi 24]
4. 5 Tenths + 63 Thousandths = _____ Thousandths [Ismailia - Fayed 24]
5. Thirty – seven and five tenths is written as _____ [Qena 24]
6. $56.98 \div 10 =$ _____ [El Beheira - Kafr El Dawar 24]
7. $2 \times$ _____ = 200,000 [Port Said 23]
8. $36.479 \approx 36.5$ [to the nearest _____] [Giza - Awseem 23]

3. Choose the correct answer.

- The place value of 8 in 85.324 is _____. [Souhag 23]
 A. Tenths. B. Tens. C. Hundredths. D. Hundreds.
- $1.5 - 0.75 =$ _____. [Alexandria - West 23]
 A. 0.75 B. 7.5 C. 1.8 D. 1.25
- 0.3  3 Thousandths. [Cairo - El Marg 23]
 A. < B. > C. =
- $34.6 \times$ _____ = 34,600 [Ismailia 24, El Beheira 23]
 A. 10 B. 100 C. 1,000 D. 10,000
- The number [fifteen and fifteen thousandths] in expanded form is _____. [Giza - El Haram 24]
 A. $10 + 5 + 0.1 + 0.005$ B. $10 + 5 + 0.05 + 0.001$
 C. $10 + 5 + 0.01 + 0.005$ D. $10 + 5 + 0.1 + 0.05$
- When divide 316 by 10, then the value of 6 becomes _____. [El Monofia - Ashmoon 24]
 A. 0.6 B. 60 C. 0.06 D. 600
- $0.02 =$ _____. [Alexandria - Agami 24]
 A. $\frac{2}{10}$ B. 2 thousandths. C. 20 thousandths. D. $\frac{20}{100}$

4. Answer the following questions.

- Decompose the number 78.096 in expanded form. [Cairo - El Sayeda Zeinab 24]

- Aya saved 17.25 pounds. Her sister saved 8.75 pounds. Find the total with them. [Souhag - Tama 24]

- Arrange from least to greatest. [Qena - Farshout 24]
 A. 0.58 B. 8.05 C. 5.8 D. 8.005

- A road of 6.975 km length, if the train travelled 2.939 km from this road,
 What is the remaining distance of the road? [Ismailia - El Kasaseen 24]
